

World Values Survey Analysis

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```
library(tidyverse)
library(car)
library(pROC)
library(caret)
library(AICcmodavg)
library(performance)
```

```
WVS <- read_csv("WVS.csv")

# cast predictors to factors
WVS$gender <- factor(WVS$gender)
WVS$religion <- factor(WVS$religion)
WVS$degree <- factor(WVS$degree)
```

```
WVS$country <- factor(WVS$country)
WVS$age <- factor(WVS$age)
```

```
# compute numbers of trials
WVS$total <- WVS$yes + WVS$no
```

```
# drop rows with 0 trials
WVS <- WVS |>
  filter(total > 0)
```

```
# compute proportions
WVS$prop <- WVS$yes/WVS$total
```

```
# ungroup data for roc-auc analysis
```

```
yes <- WVS |>
  uncount(yes) |>
  mutate(resp = 1)
```

```
no <- WVS |>
  uncount(no) |>
  mutate(resp = 0)
```

```
WVS_ungrouped <- bind_rows(yes, no) |>
  select(-c(yes, no))
```

```
rm(yes, no)
```

Model 1: logit, all predictors

1.

```
model1 <- glm(cbind(yes, no) ~ gender + religion + degree + country + age, family = binomial)
summary(model1)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age, family = binomial(link = "logit"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.47109	0.09737	4.838	1.31e-06	***
gender1	-0.19755	0.05568	-3.548	0.000388	***
religion1	-0.11143	0.08077	-1.380	0.167726	
degree1	-0.17920	0.07044	-2.544	0.010957	*
countryNorway	0.12899	0.07815	1.650	0.098848	.
countrySweden	0.44274	0.08291	5.340	9.30e-08	***
countryUSA	-0.35516	0.07353	-4.830	1.37e-06	***
age2	-0.22759	0.07520	-3.026	0.002474	**
age3	-0.22919	0.08118	-2.823	0.004754	**
age4	-0.56058	0.08326	-6.733	1.66e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.31 on 117 degrees of freedom
Residual deviance: 154.94 on 108 degrees of freedom
AIC: 580.04

Number of Fisher Scoring iterations: 3

2.

```
pchisq(deviance(model1), df.residual(model1), lower.tail = FALSE)
```

```
[1] 0.002080237
```

3.

```
model1_df = data_frame()

WVS$model1_preds <- fitted(model1, type="response")
model1_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model1_preds,
                        pred_no = WVS$total * (1 - WVS$model1_preds))
```

```
sum(model1_df$pred_yes < 5)
```

```
[1] 34
```

```
sum(model1_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model1_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model1_df$pred_no < 1)
```

```
[1] 7
```

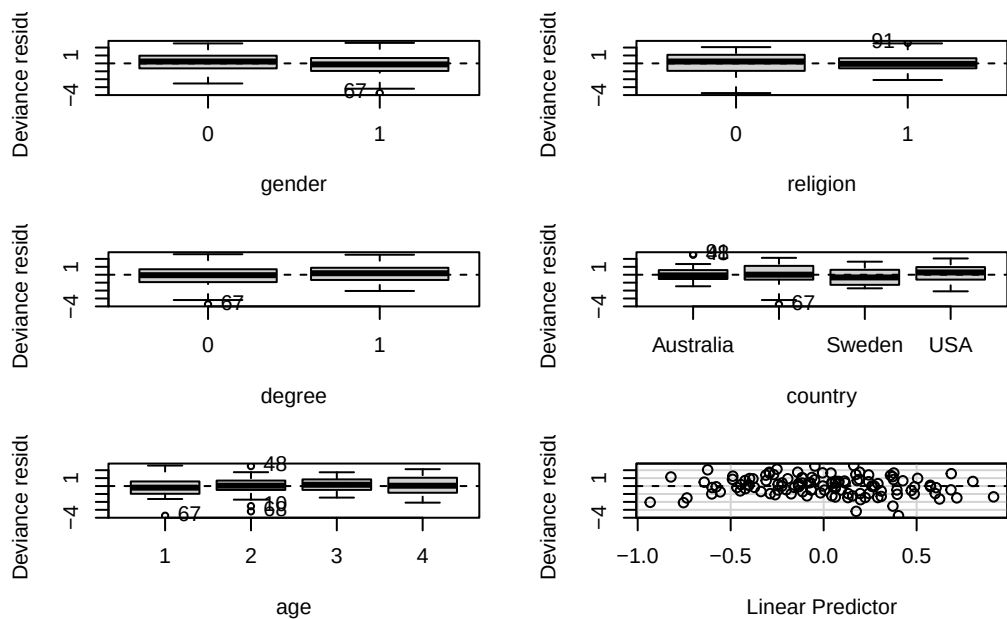
4.

```
vif(model1)
```

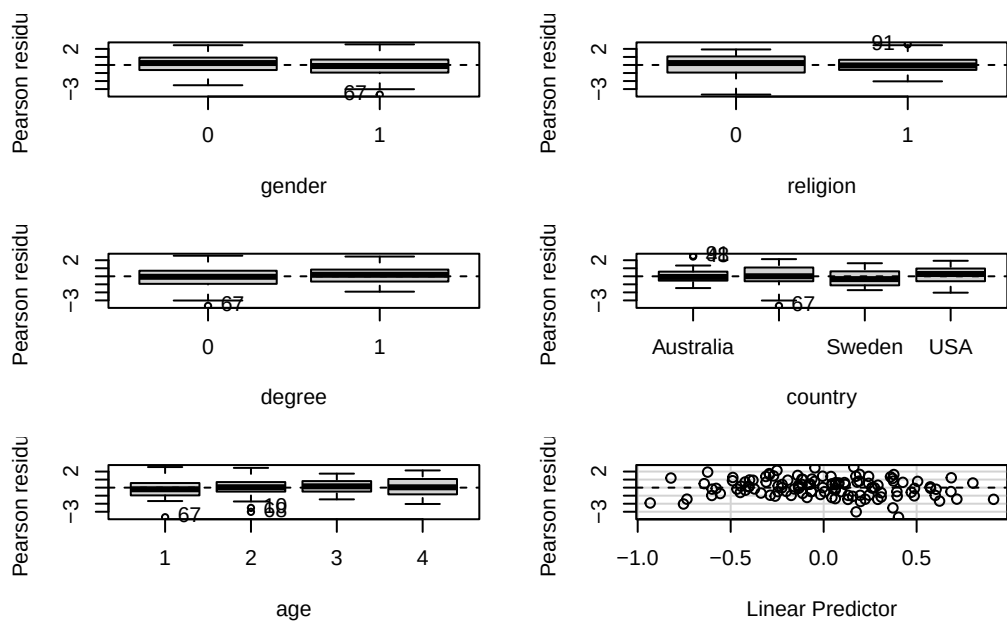
	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
gender	1.010890	1	1.005430
religion	1.067035	1	1.032974
degree	1.079023	1	1.038760
country	1.133430	3	1.021094
age	1.031296	3	1.005149

5.

```
residualPlots(model1, type = "deviance", smooth=FALSE)
```



```
residualPlots(model1, type = "pearson", smooth=FALSE)
```

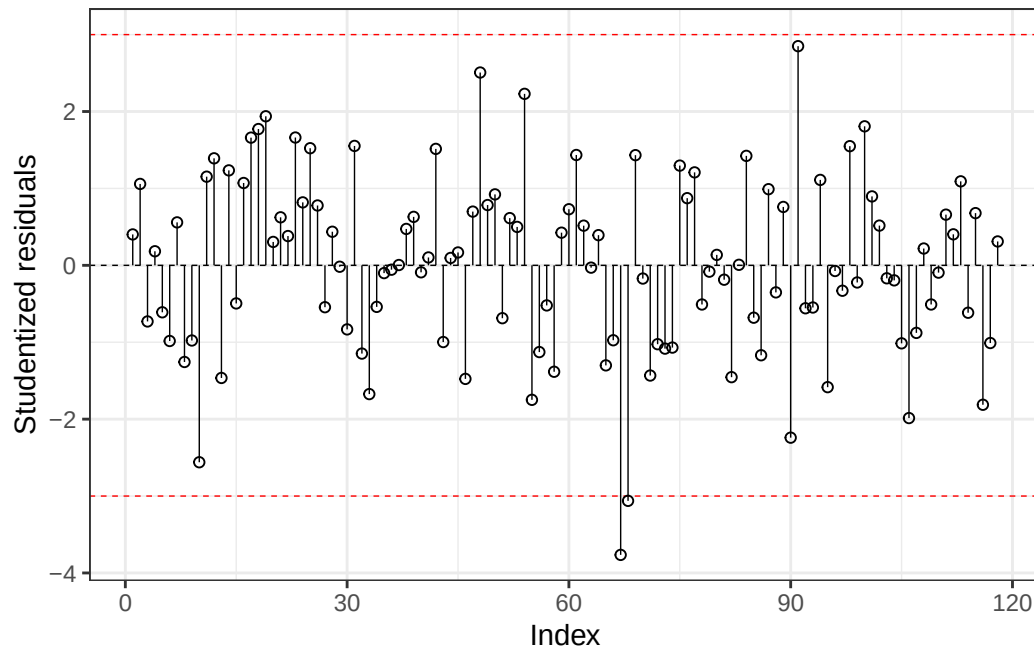


6.

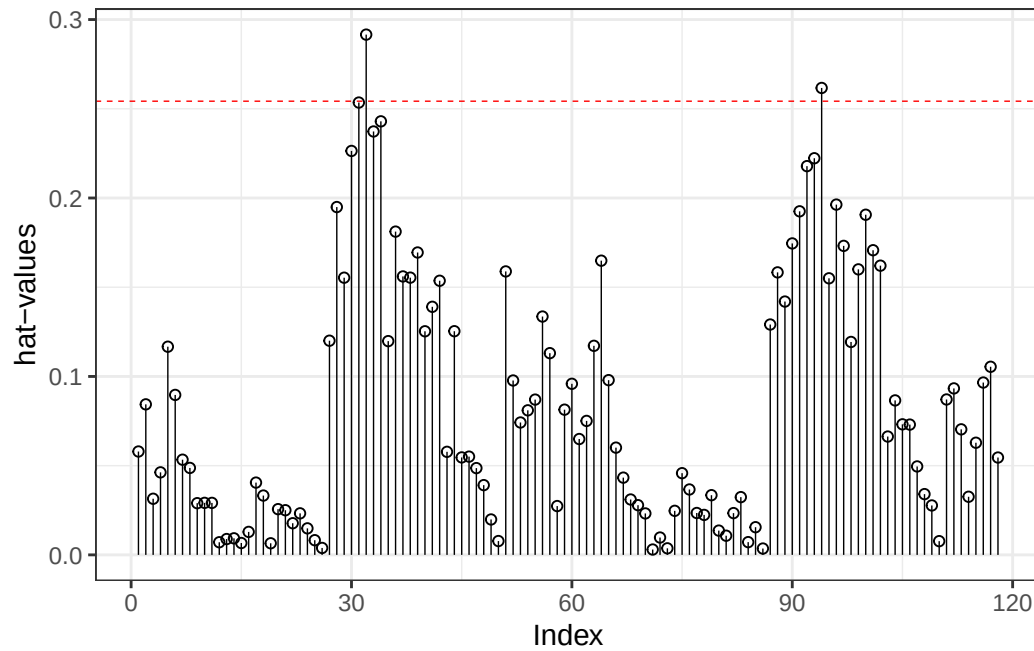
```
p <- length(coef(model1))
n <- nobs(model1)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model1_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model1),
  hat = hatvalues(model1),
  cooks = cooks.distance(model1)
)

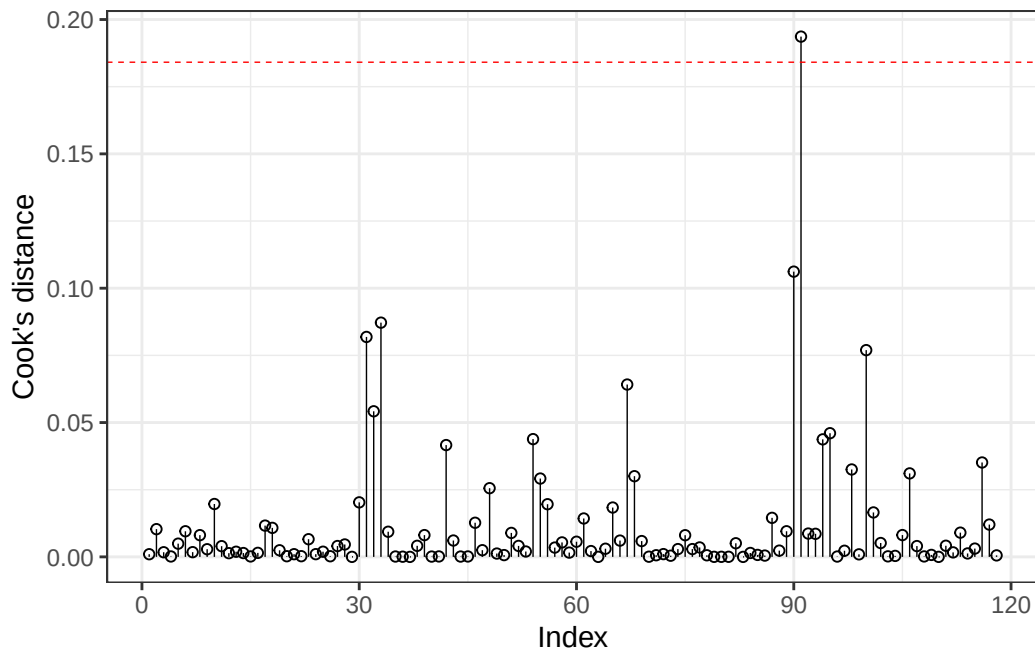
model1_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Studentized residuals") +
  theme_bw()
```



```
model1_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model1_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```

```
model1_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

	obs	studentized	hat	cooks
	32	-1.147458	0.29154998	0.05418480
	67	-3.765047	0.04329103	0.06414442
	68	-3.062337	0.03103528	0.03003680
	91	2.849746	0.19252997	0.19363523
	94	1.110984	0.26166561	0.04374303

7.

```
check_overdispersion(model1)
```

```
# Overdispersion test
```

```
dispersion ratio = 5173.822
```

```
Pearson's Chi-Squared = 558772.737
p-value = < 0.001
```

Model 2: logit, stepwise selected predictors

8.

```
full_model <- glm(cbind(yes, no) ~ (gender + religion + degree + country + age)^2, family = l
model2 <- step(full_model, direction = "both")
```

Start: AIC=566.41

```
cbind(yes, no) ~ (gender + religion + degree + country + age)^2
```

	Df	Deviance	AIC
- gender:age	3	81.856	560.95
- gender:country	3	82.422	561.52
- degree:age	3	82.988	562.08
- gender:religion	1	83.133	566.23
<none>		81.316	566.41
- gender:degree	1	83.425	566.52
- religion:age	3	88.901	568.00
- religion:degree	1	86.299	569.39
- country:age	9	103.108	570.20
- degree:country	3	93.173	572.27
- religion:country	3	96.537	575.63

Step: AIC=560.95

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + gender:country + religion:degree +
  religion:country + religion:age + degree:country + degree:age +
  country:age
```

	Df	Deviance	AIC
- gender:country	3	82.981	556.08
- degree:age	3	83.584	556.68
- gender:religion	1	83.535	560.63
- gender:degree	1	83.797	560.89
<none>		81.856	560.95
- religion:age	3	89.293	562.39

- religion:degree	1	86.818	563.91
- country:age	9	103.559	564.65
+ gender:age	3	81.316	566.41
- degree:country	3	93.782	566.88
- religion:country	3	97.140	570.23

Step: AIC=556.08

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + degree:age + country:age
```

	Df	Deviance	AIC
- degree:age	3	84.648	551.74
- gender:religion	1	84.632	555.73
<none>		82.981	556.08
- gender:degree	1	85.653	556.75
- religion:age	3	90.499	557.59
- religion:degree	1	87.894	558.99
- country:age	9	104.735	559.83
+ gender:country	3	81.856	560.95
+ gender:age	3	82.422	561.52
- degree:country	3	95.300	562.39
- religion:country	3	98.257	565.35

Step: AIC=551.74

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age
```

	Df	Deviance	AIC
- gender:religion	1	86.346	551.44
<none>		84.648	551.74
- gender:degree	1	87.318	552.41
- religion:age	3	92.017	553.11
- religion:degree	1	89.855	554.95
+ degree:age	3	82.981	556.08
+ gender:country	3	83.584	556.68
+ gender:age	3	84.030	557.12
- degree:country	3	96.308	557.40
- country:age	9	109.099	558.19
- religion:country	3	99.803	560.90

Step: AIC=551.44

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:degree + religion:degree + religion:country + religion:age +
  degree:country + country:age
```

	Df	Deviance	AIC
<none>		86.346	551.44
+ gender:religion	1	84.648	551.74
- gender:degree	1	88.972	552.07
- religion:age	3	93.481	552.58
- religion:degree	1	91.534	554.63
+ degree:age	3	84.632	555.73
+ gender:country	3	85.312	556.41
+ gender:age	3	85.850	556.94
- degree:country	3	97.978	557.07
- country:age	9	111.007	558.10
- religion:country	3	101.334	560.43

```
rm(full_model)

summary(model2)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
  age + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age, family = binomial(link = "logit"),
  data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.217336	0.164491	1.321	0.186411
gender1	-0.148869	0.063207	-2.355	0.018511 *
religion1	0.413382	0.174472	2.369	0.017820 *
degree1	0.464299	0.245440	1.892	0.058531 .
countryNorway	-0.772140	0.265397	-2.909	0.003622 **
countrySweden	-0.506866	0.552069	-0.918	0.358555
countryUSA	-0.132150	0.212675	-0.621	0.534356
age2	-0.192303	0.199186	-0.965	0.334322
age3	-0.078153	0.231606	-0.337	0.735784
age4	-0.481729	0.256649	-1.877	0.060519 .
gender1:degree1	-0.223754	0.138193	-1.619	0.105417
religion1:degree1	-0.481197	0.211773	-2.272	0.023073 *

religion1:countryNorway	0.339009	0.242029	1.401	0.161305	
religion1:countrySweden	0.723014	0.541248	1.336	0.181606	
religion1:countryUSA	-0.471010	0.183692	-2.564	0.010344	*
religion1:age2	-0.346354	0.206346	-1.679	0.093247	.
religion1:age3	-0.617478	0.235896	-2.618	0.008855	**
religion1:age4	-0.373137	0.256673	-1.454	0.146018	
degree1:countryNorway	0.176913	0.225390	0.785	0.432501	
degree1:countrySweden	-0.452021	0.237088	-1.907	0.056578	.
degree1:countryUSA	-0.238777	0.221669	-1.077	0.281401	
countryNorway:age2	0.494555	0.204188	2.422	0.015433	*
countrySweden:age2	0.377771	0.223872	1.687	0.091518	.
countryUSA:age2	0.313921	0.202611	1.549	0.121290	
countryNorway:age3	0.812973	0.221695	3.667	0.000245	***
countrySweden:age3	0.530452	0.233264	2.274	0.022964	*
countryUSA:age3	0.420380	0.223193	1.883	0.059635	.
countryNorway:age4	0.791850	0.232416	3.407	0.000657	***
countrySweden:age4	0.431749	0.247510	1.744	0.081095	.
countryUSA:age4	-0.007768	0.220883	-0.035	0.971945	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 86.346 on 88 degrees of freedom
 AIC: 551.44

Number of Fisher Scoring iterations: 4

9.

```
pchisq(deviance(model2), df.residual(model2), lower.tail = FALSE)
```

```
[1] 0.5299432
```

10.

```

model2_df = data_frame()

WVS$model2_preds <- fitted(model2, type="response")
model2_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model2_preds,
                        pred_no = WVS$total * (1 - WVS$model2_preds))

sum(model2_df$pred_yes < 5)

```

[1] 36

```
sum(model2_df$pred_yes < 1)
```

[1] 6

```
sum(model2_df$pred_no < 5)
```

[1] 33

```
sum(model2_df$pred_no < 1)
```

[1] 7

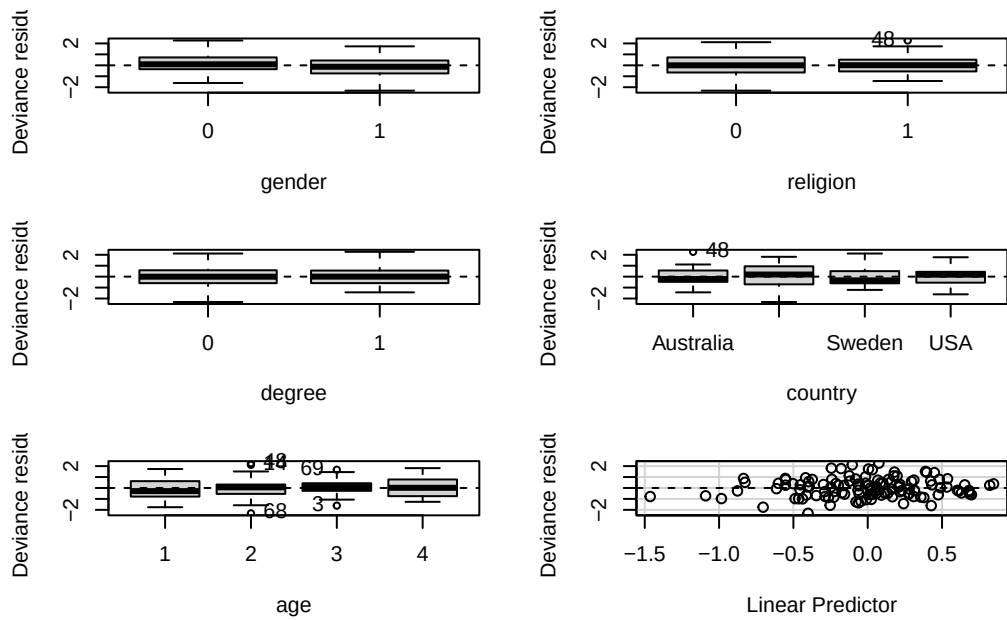
11.

```
vif(model2)
```

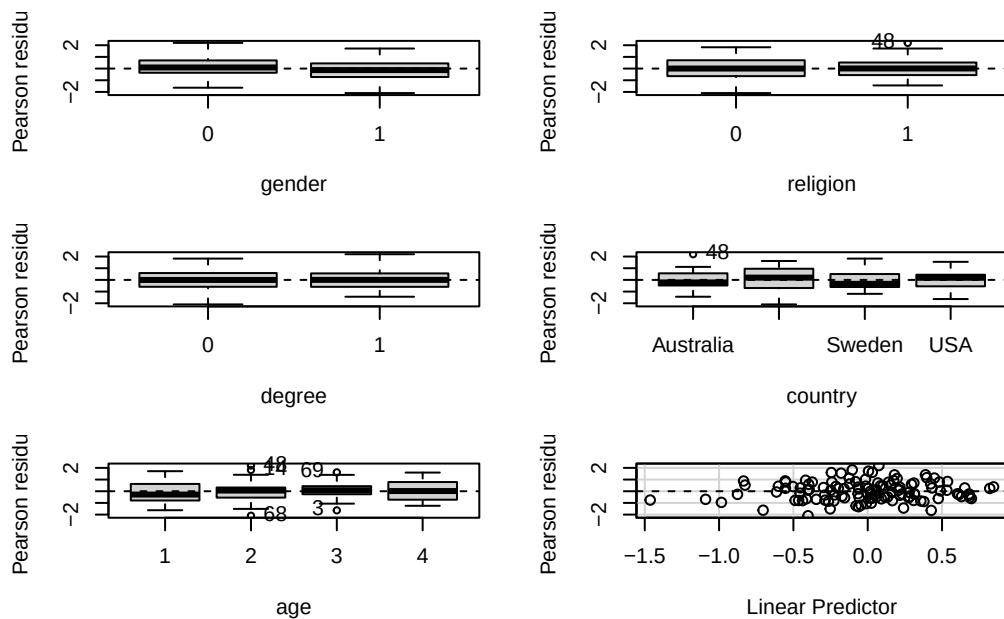
	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
gender	1.286506	1	1.134243
religion	4.983209	1	2.232310
degree	12.887581	1	3.589928
country	6214.764574	3	4.287829
age	655.066666	3	2.947005
gender:degree	2.193618	1	1.481087
religion:degree	8.548454	1	2.923774
religion:country	3100.137170	3	3.818535
religion:age	644.190630	3	2.938793
degree:country	16.572878	3	1.596736
country:age	1121.349819	9	1.477169

12.

```
residualPlots(model2, type = "deviance", smooth=FALSE)
```



```
residualPlots(model2, type = "pearson", smooth=FALSE)
```



13.

```
p <- length(coef(model2))
n <- nobs(model2)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model2_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model2),
  hat = hatvalues(model2),
  cooks = cooks.distance(model2)
)

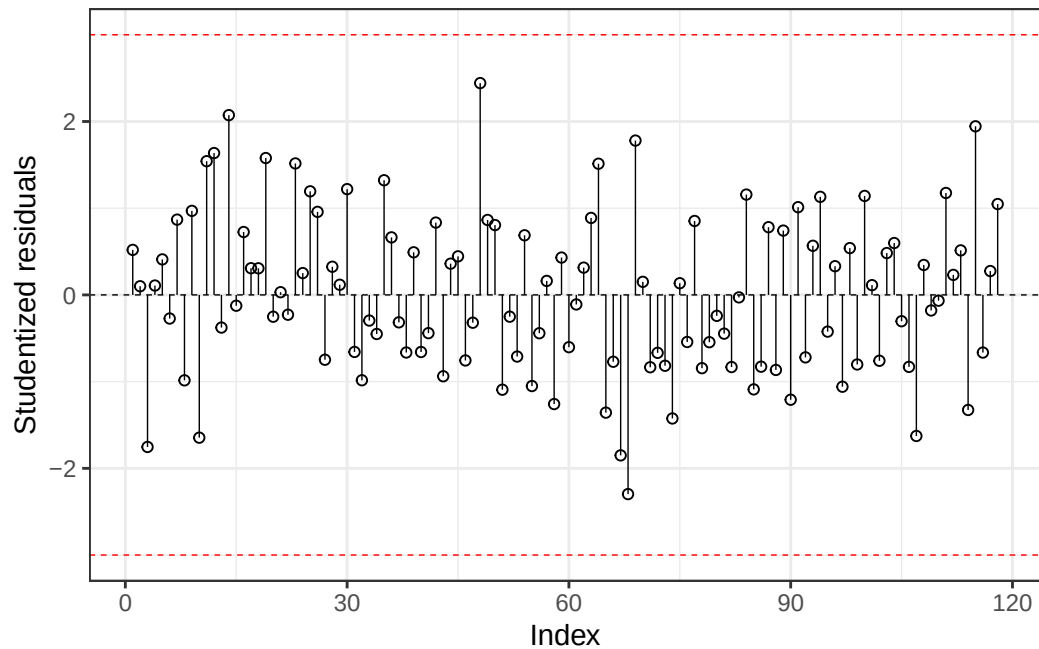
model2_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```



```

xlab("Index") +
ylab("Studentized residuals") +
theme_bw()

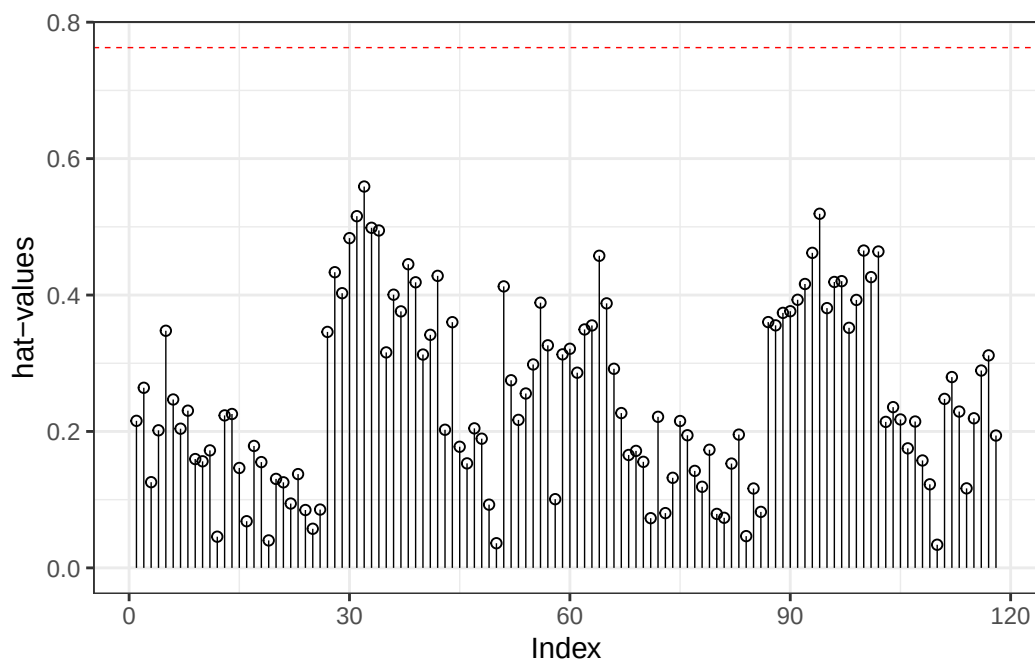
```



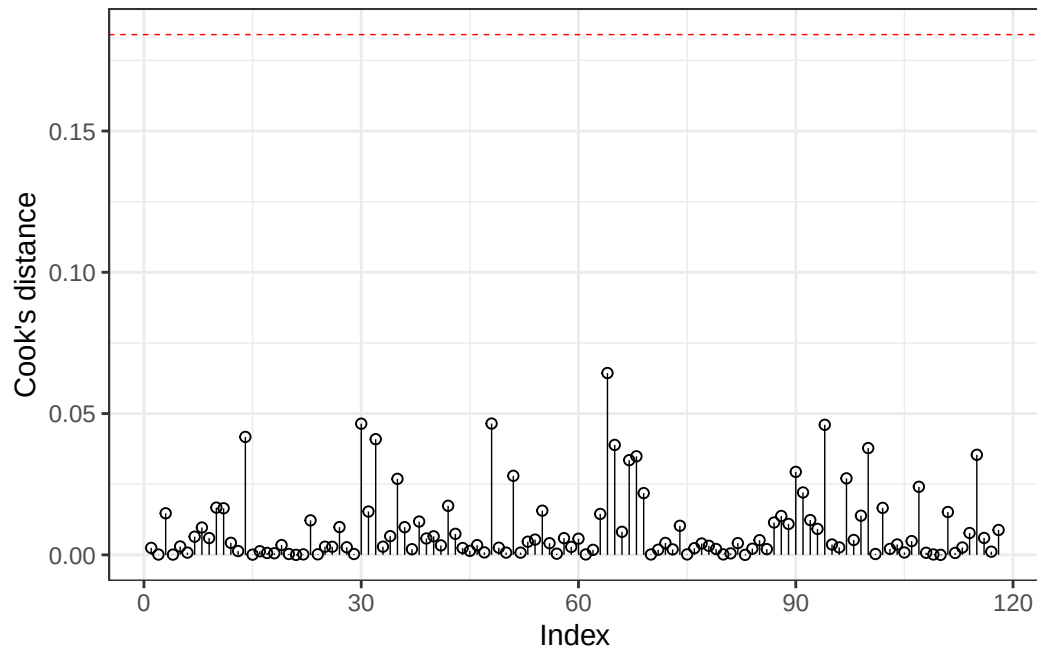
```

model2_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()

```



```
model2_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model2_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

14.

```
check_overdispersion(model2)
```

```
# Overdispersion test
```

```
dispersion ratio = 6492.736
Pearson's Chi-Squared = 571360.766
p-value = < 0.001
```

Model 3: probit, all predictors

15.

```
model3 <- glm(cbind(yes, no) ~ gender + religion + degree + country + age, family = binomial)
summary(model3)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age, family = binomial(link = "probit"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.29411	0.06059	4.854	1.21e-06	***
gender1	-0.12329	0.03466	-3.557	0.000375	***
religion1	-0.06984	0.05034	-1.387	0.165296	
degree1	-0.11221	0.04380	-2.562	0.010416	*
countryNorway	0.08029	0.04881	1.645	0.099996	.
countrySweden	0.27581	0.05150	5.356	8.52e-08	***
countryUSA	-0.22242	0.04578	-4.858	1.19e-06	***
age2	-0.14139	0.04683	-3.019	0.002535	**
age3	-0.14234	0.05055	-2.815	0.004870	**
age4	-0.34997	0.05174	-6.764	1.34e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.31 on 117 degrees of freedom
Residual deviance: 154.70 on 108 degrees of freedom
AIC: 579.79

Number of Fisher Scoring iterations: 4

16.

```
pchisq(deviance(model3), df.residual(model3), lower.tail = FALSE)
```

```
[1] 0.002168671
```

17.

```
model3_df = data_frame()

WVS$model3_preds <- fitted(model3, type="response")
model3_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model3_preds,
                        pred_no = WVS$total * (1 - WVS$model3_preds))

sum(model3_df$pred_yes < 5)
```

```
[1] 34
```

```
sum(model3_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model3_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model3_df$pred_no < 1)
```

```
[1] 7
```

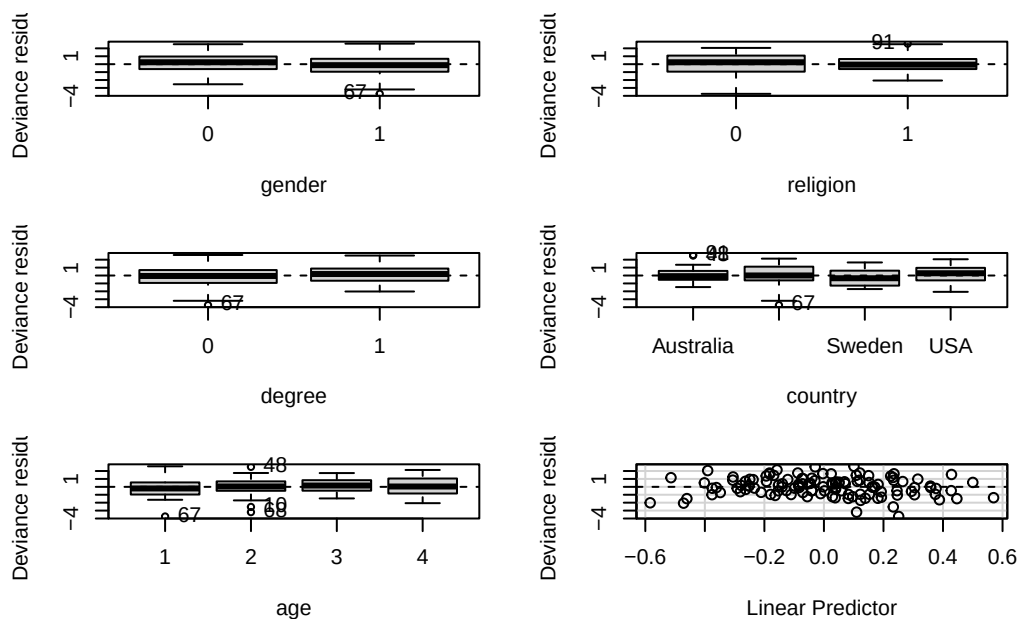
18.

```
vif(model3)
```

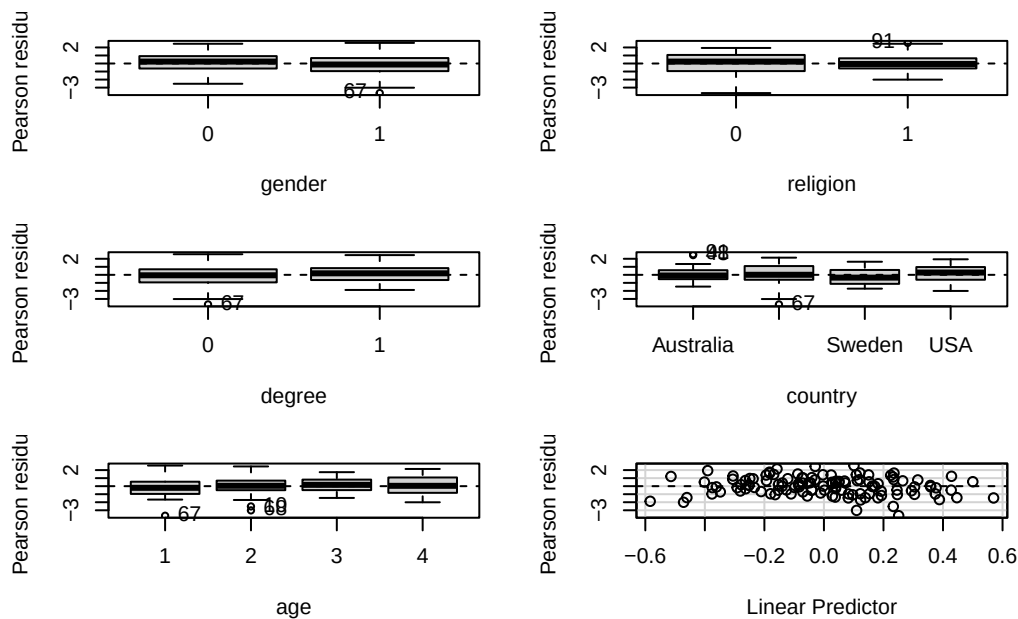
	GVIF	Df	GVIF ^{1/(2*Df)}
gender	1.010596	1	1.005284
religion	1.066794	1	1.032857
degree	1.077808	1	1.038175
country	1.134183	3	1.021207
age	1.032859	3	1.005403

19.

```
residualPlots(model3, type = "deviance", smooth=FALSE)
```



```
residualPlots(model3, type = "pearson", smooth=FALSE)
```



20.

```
p <- length(coef(model3))
n <- nobs(model3)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

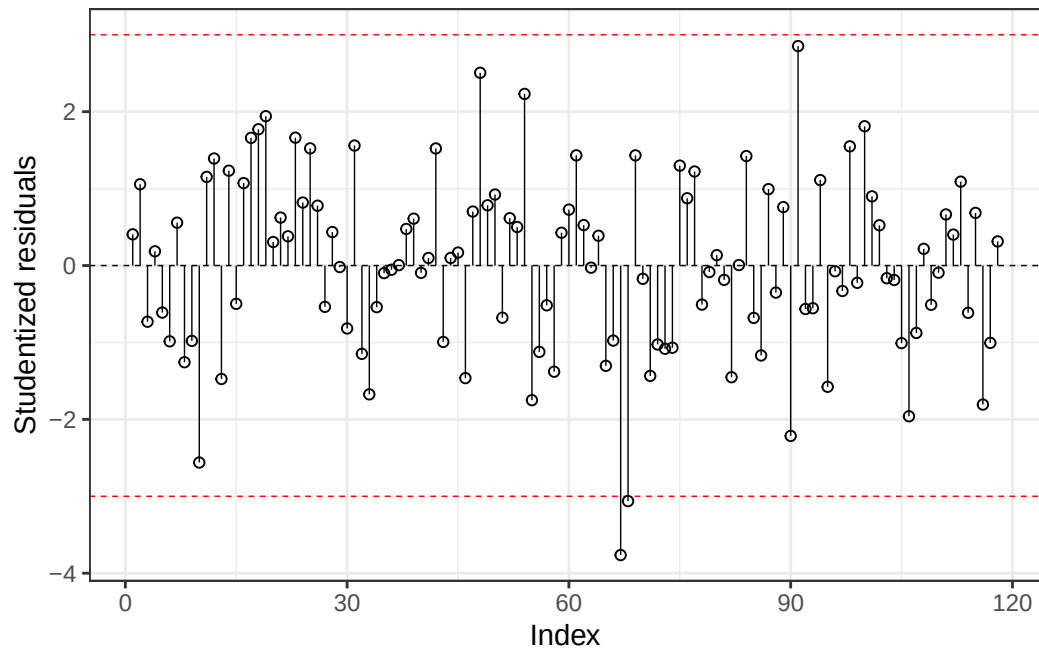
model3_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model3),
  hat = hatvalues(model3),
  cooks = cooks.distance(model3)
)

model3_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

```

xlab("Index") +
ylab("Studentized residuals") +
theme_bw()

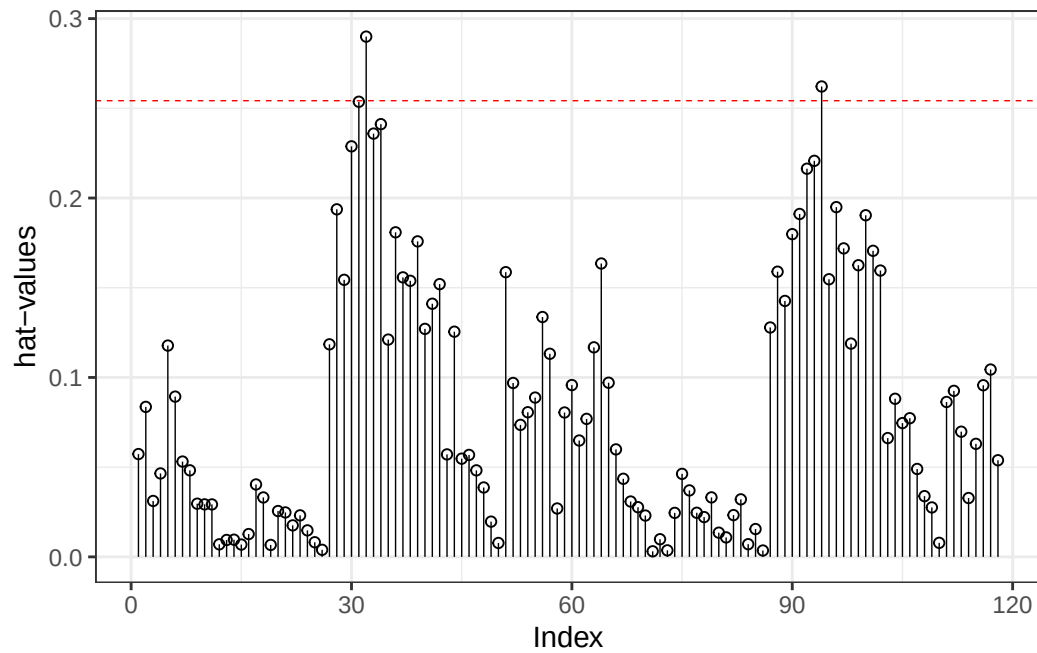
```



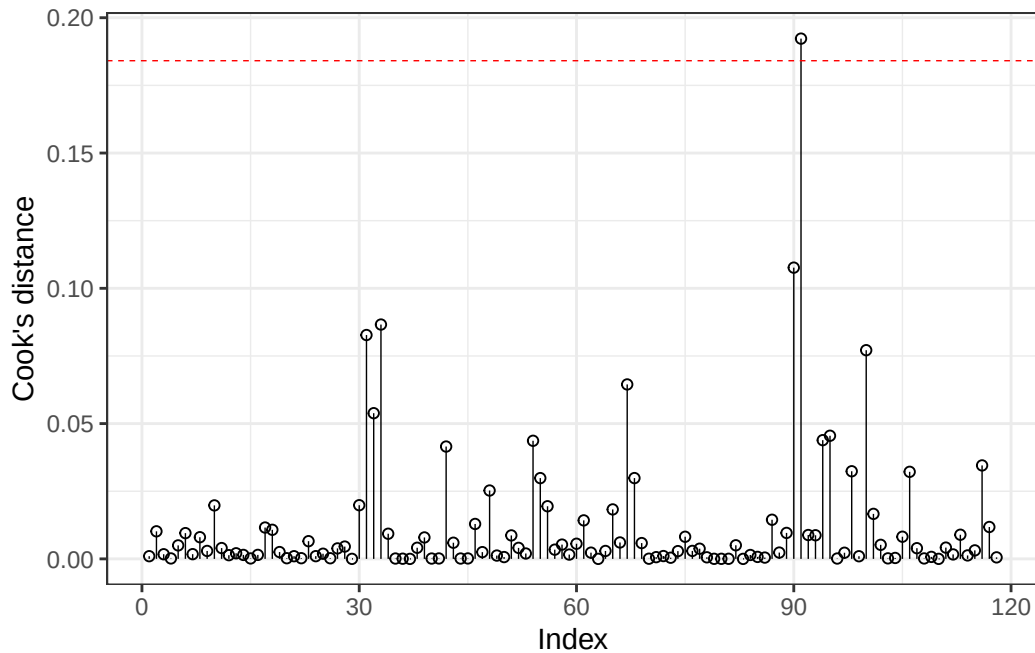
```

model3_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()

```

```
model3_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model3_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

	obs	studentized	hat	cooks
	32	-1.148298	0.28995901	0.05384715
	67	-3.763985	0.04352978	0.06447789
	68	-3.062507	0.03085690	0.02986197
	91	2.852854	0.19110592	0.19228324
	94	1.111268	0.26216568	0.04387877

21.

```
check_overdispersion(model3)
```

```
# Overdispersion test
```

```
dispersion ratio = 5175.908
```

```
Pearson's Chi-Squared = 558998.029
p-value = < 0.001
```

Model 4: probit, stepwise selected predictors

22.

```
full_model <- glm(cbind(yes, no) ~ (gender + religion + degree + country + age)^2, family = l
model4 <- step(full_model, direction = "both")
```

Start: AIC=566.14

```
cbind(yes, no) ~ (gender + religion + degree + country + age)^2
```

	Df	Deviance	AIC
- gender:age	3	81.577	560.67
- gender:country	3	82.166	561.26
- degree:age	3	82.697	561.79
- gender:religion	1	82.892	565.99
<none>		81.046	566.14
- gender:degree	1	83.115	566.21
- religion:age	3	88.641	567.74
- religion:degree	1	86.074	569.17
- country:age	9	102.967	570.06
- degree:country	3	92.828	571.92
- religion:country	3	96.265	575.36

Step: AIC=560.67

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + gender:country + religion:degree +
  religion:country + religion:age + degree:country + degree:age +
  country:age
```

	Df	Deviance	AIC
- gender:country	3	82.722	555.82
- degree:age	3	83.284	556.38
- gender:religion	1	83.282	560.38
- gender:degree	1	83.483	560.58
<none>		81.577	560.67
- religion:age	3	89.032	562.13

- religion:degree	1	86.582	563.68
- country:age	9	103.416	564.51
+ gender:age	3	81.046	566.14
- degree:country	3	93.434	566.53
- religion:country	3	96.860	569.96

Step: AIC=555.82

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + degree:age + country:age
```

	Df	Deviance	AIC
- degree:age	3	84.371	551.47
- gender:religion	1	84.392	555.49
<none>		82.722	555.82
- gender:degree	1	85.367	556.46
- religion:age	3	90.257	557.35
- religion:degree	1	87.672	558.77
- country:age	9	104.600	559.69
+ gender:country	3	81.577	560.67
+ gender:age	3	82.166	561.26
- degree:country	3	94.982	562.08
- religion:country	3	97.994	565.09

Step: AIC=551.47

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age
```

	Df	Deviance	AIC
- gender:religion	1	86.087	551.18
<none>		84.371	551.47
- gender:degree	1	87.020	552.11
- religion:age	3	91.763	552.86
- religion:degree	1	89.624	554.72
+ degree:age	3	82.722	555.82
+ gender:country	3	83.284	556.38
+ gender:age	3	83.753	556.85
- degree:country	3	95.992	557.09
- country:age	9	108.913	558.01
- religion:country	3	99.518	560.61

Step: AIC=551.18

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:degree + religion:degree + religion:country + religion:age +
  degree:country + country:age
```

	Df	Deviance	AIC
<none>		86.087	551.18
+ gender:religion	1	84.371	551.47
- gender:degree	1	88.696	551.79
- religion:age	3	93.224	552.32
- religion:degree	1	91.313	554.41
+ degree:age	3	84.392	555.49
+ gender:country	3	85.036	556.13
+ gender:age	3	85.593	556.69
- degree:country	3	97.675	556.77
- country:age	9	110.836	557.93
- religion:country	3	101.041	560.14

```
rm(full_model)
```

```
summary(model4)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
  age + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age, family = binomial(link = "probit"),
  data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.1353176	0.1026749	1.318	0.187529
gender1	-0.0929651	0.0392732	-2.367	0.017926 *
religion1	0.2577086	0.1086782	2.371	0.017726 *
degree1	0.2886872	0.1522160	1.897	0.057886 .
countryNorway	-0.4829571	0.1647993	-2.931	0.003383 **
countrySweden	-0.3198489	0.3444029	-0.929	0.353042
countryUSA	-0.0830294	0.1326532	-0.626	0.531372
age2	-0.1194715	0.1243734	-0.961	0.336760
age3	-0.0481424	0.1445296	-0.333	0.739061
age4	-0.3019042	0.1595362	-1.892	0.058440 .
gender1:degree1	-0.1383816	0.0857054	-1.615	0.106393
religion1:degree1	-0.3005951	0.1314340	-2.287	0.022193 *

```

religion1:countryNorway  0.2130327  0.1505351   1.415 0.157019
religion1:countrySweden  0.4524676  0.3378559   1.339 0.180495
religion1:countryUSA     -0.2928562  0.1145436  -2.557 0.010566 *
religion1:age2           -0.2160117  0.1287079  -1.678 0.093287 .
religion1:age3           -0.3856543  0.1470353  -2.623 0.008719 **
religion1:age4           -0.2311286  0.1593588  -1.450 0.146956
degree1:countryNorway    0.1113984  0.1403285   0.794 0.427289
degree1:countrySweden    -0.2799168  0.1473622  -1.900 0.057497 .
degree1:countryUSA       -0.1464712  0.1372303  -1.067 0.285819
countryNorway:age2       0.3082251  0.1272573   2.422 0.015433 *
countrySweden:age2       0.2369752  0.1382451   1.714 0.086498 .
countryUSA:age2          0.1954973  0.1262616   1.548 0.121538
countryNorway:age3       0.5074481  0.1380978   3.675 0.000238 ***
countrySweden:age3       0.3323004  0.1441206   2.306 0.021127 *
countryUSA:age3          0.2618948  0.1389375   1.885 0.059432 .
countryNorway:age4       0.4945015  0.1448566   3.414 0.000641 ***
countrySweden:age4       0.2722940  0.1530869   1.779 0.075291 .
countryUSA:age4         -0.0006463  0.1362876  -0.005 0.996216
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 319.307  on 117  degrees of freedom
Residual deviance: 86.087  on 88  degrees of freedom
AIC: 551.18

```

Number of Fisher Scoring iterations: 4

23.

```
pchisq(deviance(model4), df.residual(model4), lower.tail = FALSE)
```

```
[1] 0.5377944
```

24.

```

model4_df = data_frame()

WVS$model4_preds <- fitted(model4, type="response")
model4_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model4_preds,
                        pred_no = WVS$total * (1 - WVS$model4_preds))

sum(model4_df$pred_yes < 5)

```

[1] 36

```
sum(model4_df$pred_yes < 1)
```

[1] 6

```
sum(model4_df$pred_no < 5)
```

[1] 33

```
sum(model4_df$pred_no < 1)
```

[1] 7

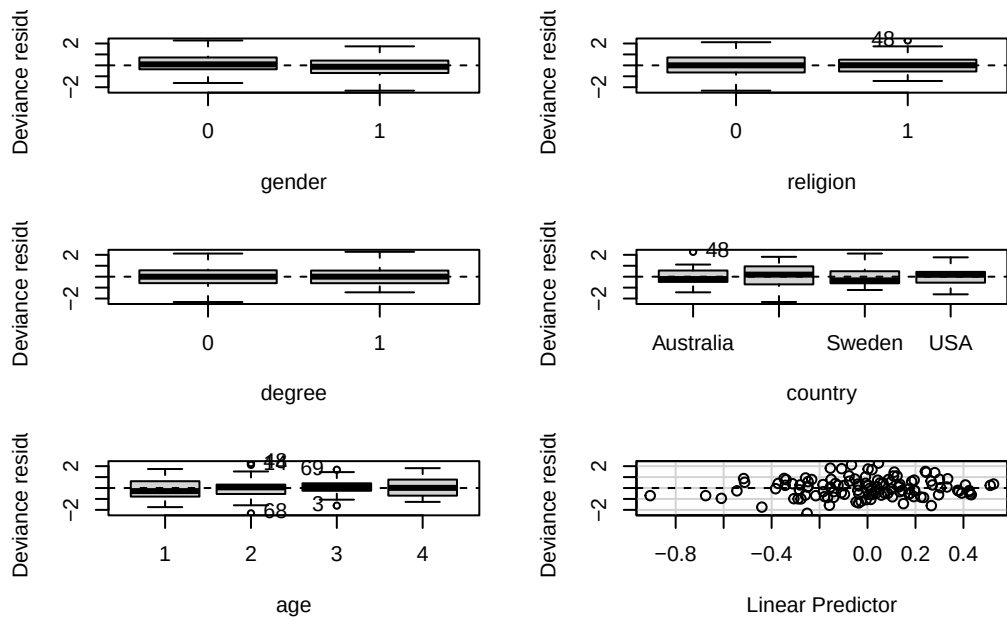
25.

```
vif(model4)
```

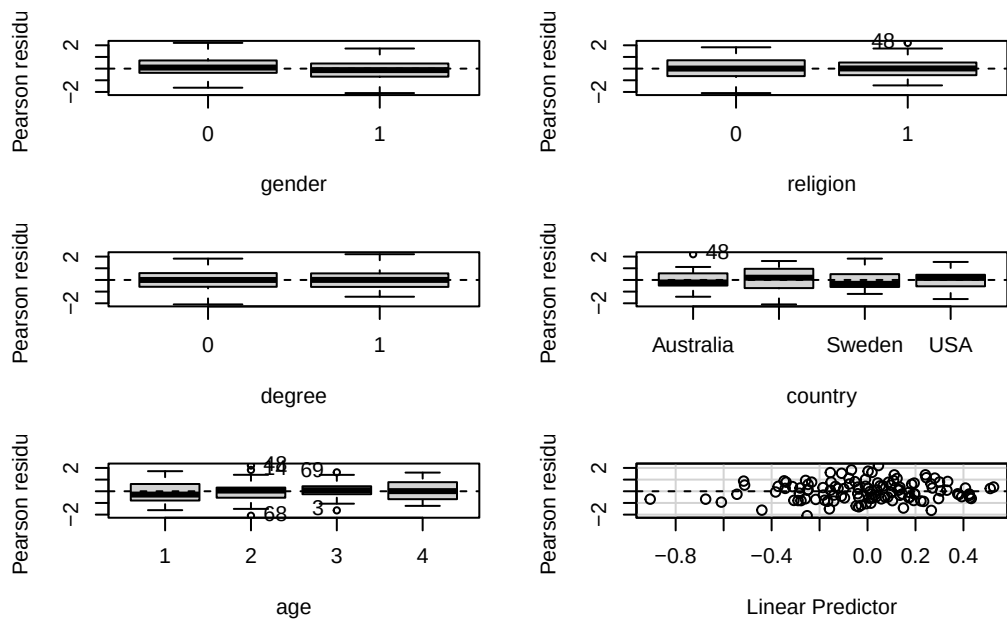
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.287390	1	1.134632
religion	4.974733	1	2.230411
degree	12.880581	1	3.588953
country	6343.949938	3	4.302557
age	671.535394	3	2.959226
gender:degree	2.199001	1	1.482903
religion:degree	8.563759	1	2.926390
religion:country	3191.480009	3	3.837061
religion:age	655.360564	3	2.947226
degree:country	16.560507	3	1.596537
country:age	1143.118337	9	1.478747

26.

```
residualPlots(model4, type = "deviance", smooth=FALSE)
```



```
residualPlots(model4, type = "pearson", smooth=FALSE)
```

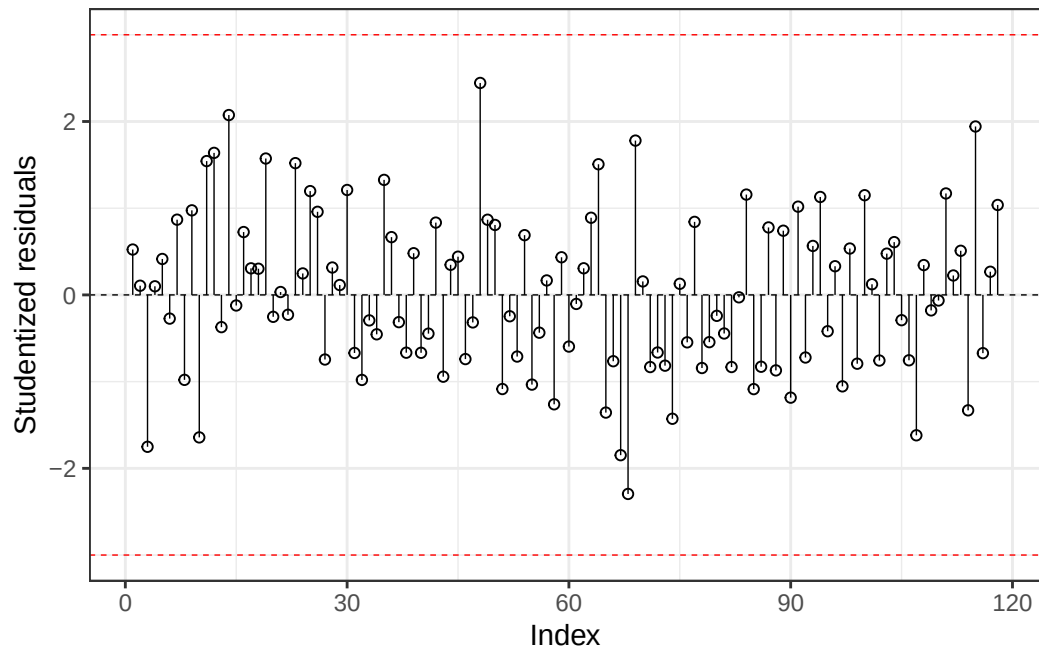
27.

```
p <- length(coef(model4))
n <- nobs(model4)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

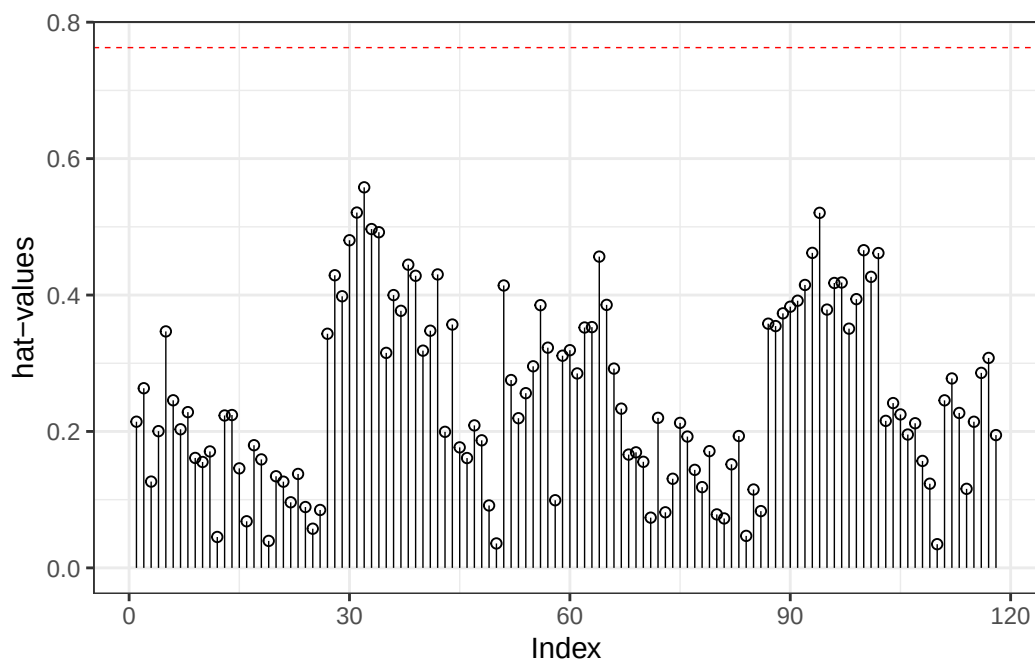
model4_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model4),
  hat = hatvalues(model4),
  cooks = cooks.distance(model4)
)

model4_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

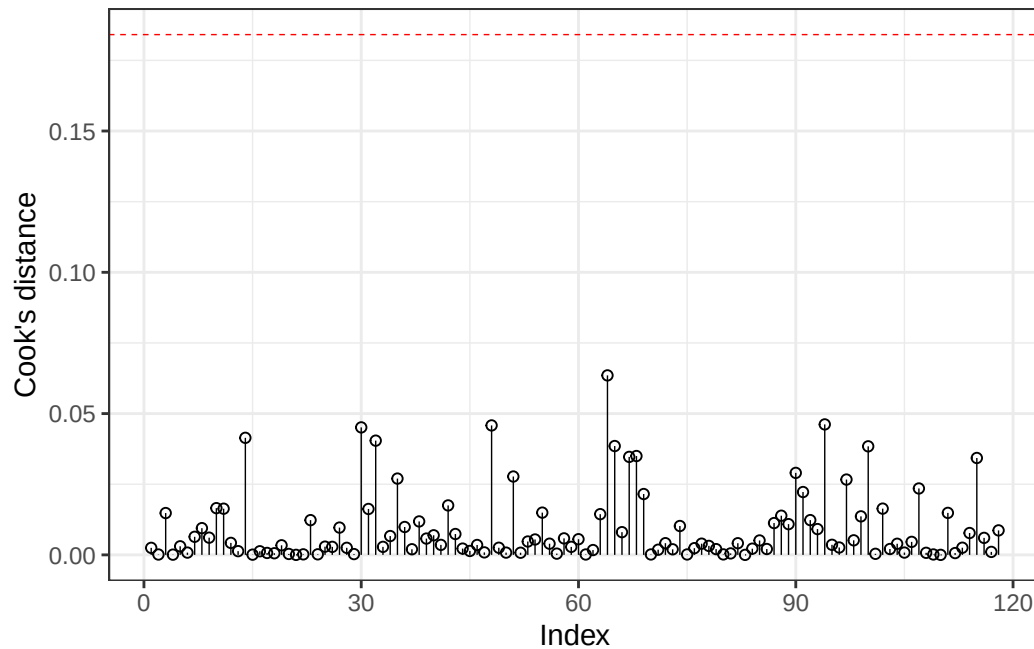
```
xlab("Index") +
ylab("Studentized residuals") +
theme_bw()
```



```
model4_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model4_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model4_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

28.

```
check_overdispersion(model4)
```

```
# Overdispersion test
```

```
dispersion ratio = 6495.589
Pearson's Chi-Squared = 571611.845
p-value = < 0.001
```

Model 5: cloglog, all predictors

29.

```
model5 <- glm(cbind(yes, no) ~ gender + religion + degree + country + age, family = binomial)
summary(model5)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age, family = binomial(link = "cloglog"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.05051	0.06763	-0.747	0.455212
gender1	-0.13779	0.03964	-3.476	0.000509 ***
religion1	-0.06455	0.05803	-1.112	0.265967
degree1	-0.13029	0.05056	-2.577	0.009974 **
countryNorway	0.08391	0.05509	1.523	0.127747
countrySweden	0.30004	0.05599	5.359	8.38e-08 ***
countryUSA	-0.26406	0.05508	-4.794	1.63e-06 ***
age2	-0.15620	0.05186	-3.012	0.002595 **
age3	-0.15710	0.05607	-2.802	0.005083 **
age4	-0.39704	0.06007	-6.610	3.84e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.31 on 117 degrees of freedom
Residual deviance: 157.33 on 108 degrees of freedom
AIC: 582.42

Number of Fisher Scoring iterations: 5

30.

```
pchisq(deviance(model5), df.residual(model5), lower.tail = FALSE)
```

```
[1] 0.001380655
```

31.

```
model5_df = data_frame()

WVS$model5_preds <- fitted(model5, type="response")
model5_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model5_preds,
                        pred_no = WVS$total * (1 - WVS$model5_preds))

sum(model5_df$pred_yes < 5)
```

```
[1] 34
```

```
sum(model5_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model5_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model5_df$pred_no < 1)
```

```
[1] 7
```

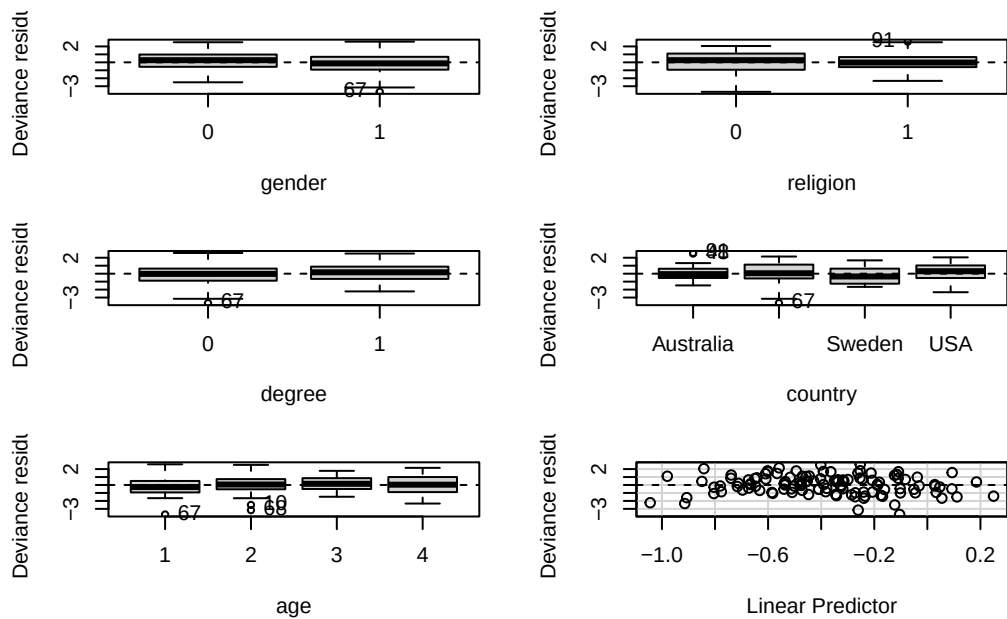
32.

```
vif(model5)
```

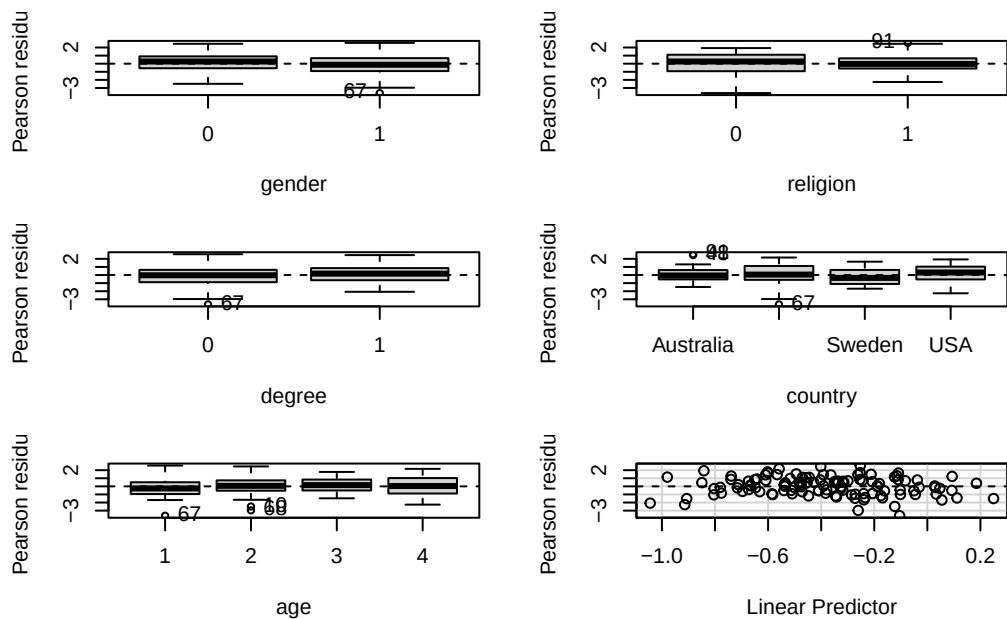
	GVIF	Df	GVIF ^{1/(2*Df)}
gender	1.010499	1	1.005236
religion	1.073340	1	1.036021
degree	1.074092	1	1.036384
country	1.139045	3	1.021935
age	1.030564	3	1.005030

33.

```
residualPlots(model5, type = "deviance", smooth=FALSE)
```



```
residualPlots(model5, type = "pearson", smooth=FALSE)
```



34.

```
p <- length(coef(model5))
n <- nobs(model5)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model5_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model5),
  hat = hatvalues(model5),
  cooks = cooks.distance(model5)
)

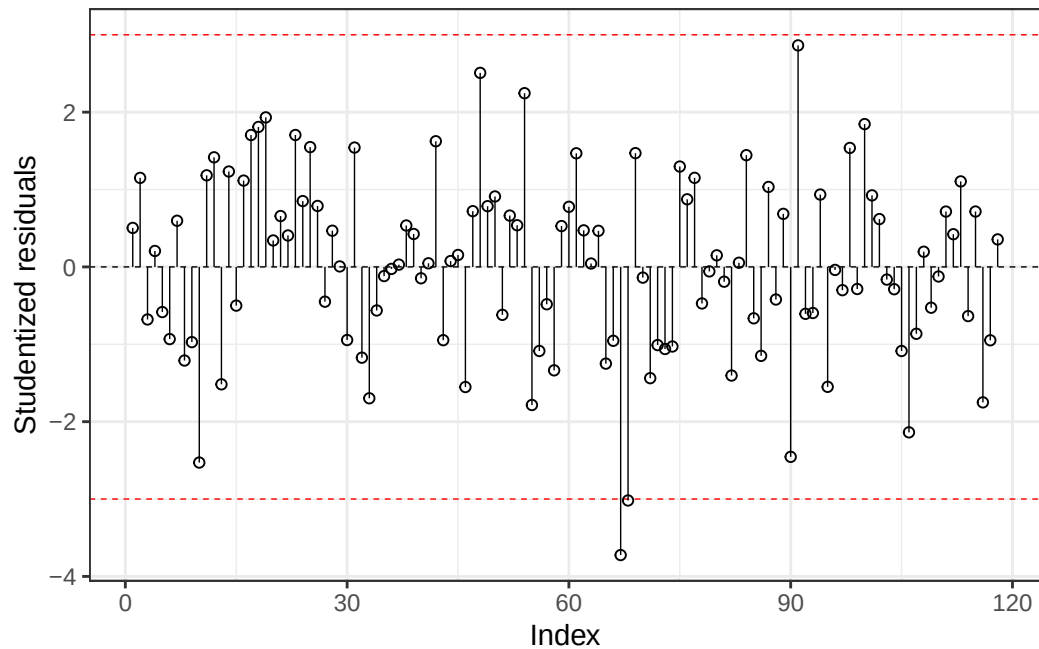
model5_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```



```

xlab("Index") +
ylab("Studentized residuals") +
theme_bw()

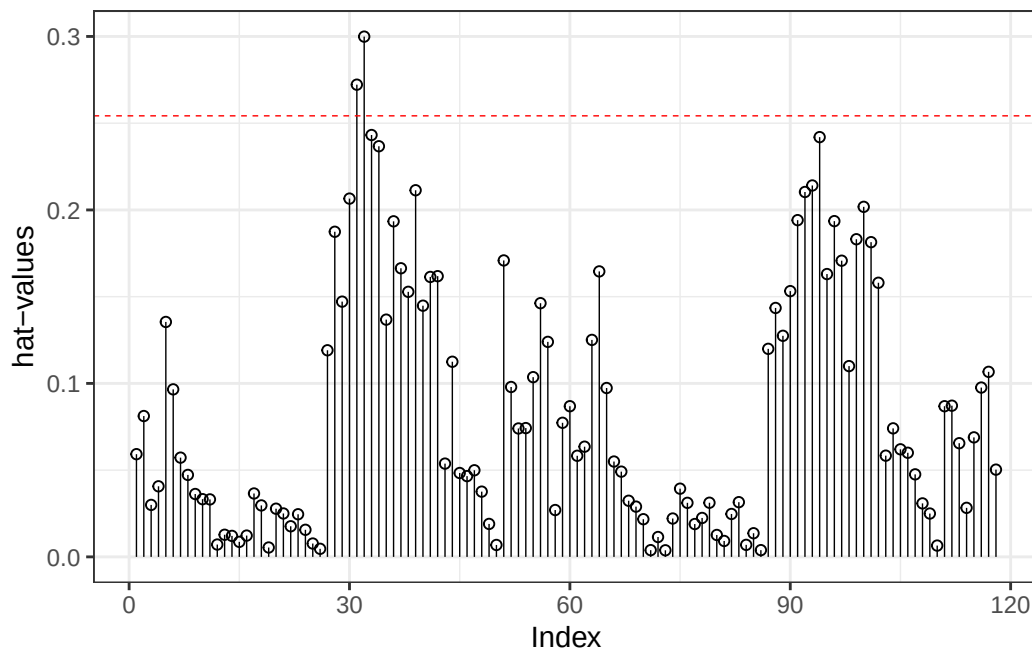
```



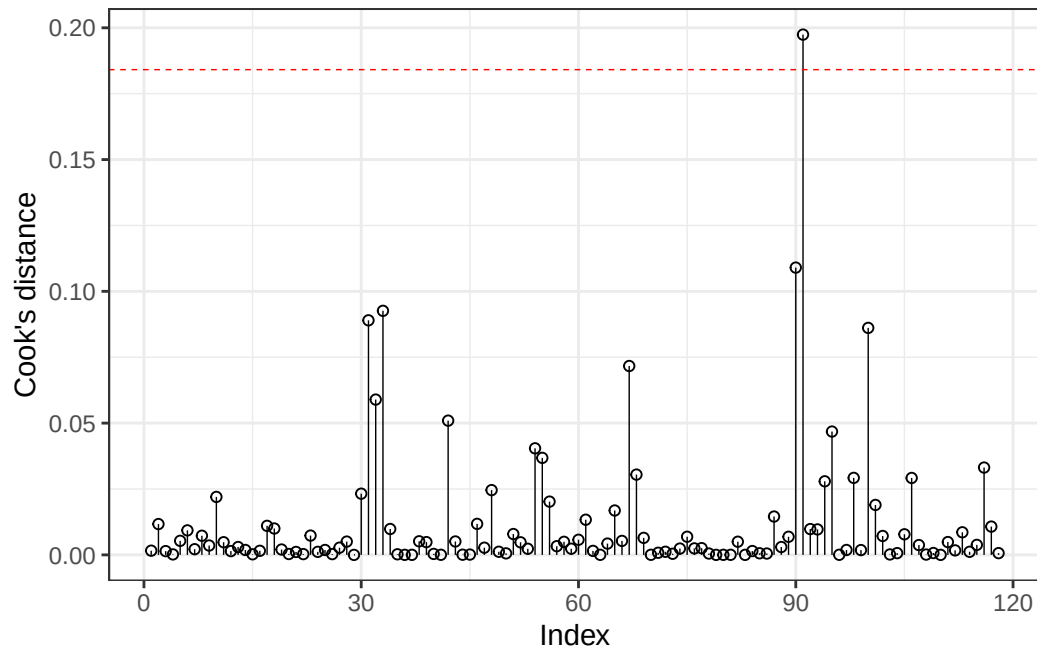
```

model5_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()

```



```
model5_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model5_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

	obs	studentized	hat	cooks
	31	1.542559	0.27221056	0.08899846
	32	-1.172895	0.29990446	0.05893100
	67	-3.723762	0.04914319	0.07166579
	68	-3.018284	0.03236015	0.03046611
	91	2.862841	0.19411139	0.19741059

35.

```
check_overdispersion(model5)
```

```
# Overdispersion test
```

```
dispersion ratio = 5154.763
```

```
Pearson's Chi-Squared = 556714.398
p-value = < 0.001
```

Model 6: cloglog, stepwise selected predictors

36.

```
full_model <- glm(cbind(yes, no) ~ (gender + religion + degree + country + age)^2, family = l
model6 <- step(full_model, direction = "both")
```

Start: AIC=568.04

```
cbind(yes, no) ~ (gender + religion + degree + country + age)^2
```

	Df	Deviance	AIC
- gender:age	3	83.558	562.65
- gender:country	3	83.965	563.06
- degree:age	3	84.455	563.55
- gender:religion	1	84.582	567.68
<none>		82.943	568.04
- gender:degree	1	85.198	568.29
- religion:age	3	90.128	569.22
- religion:degree	1	88.070	571.16
- degree:country	3	94.316	573.41
- country:age	9	106.403	573.50
- religion:country	3	97.304	576.40

Step: AIC=562.65

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + gender:country + religion:degree +
  religion:country + religion:age + degree:country + degree:age +
  country:age
```

	Df	Deviance	AIC
- gender:country	3	84.643	557.74
- degree:age	3	85.145	558.24
- gender:religion	1	85.083	562.18
<none>		83.558	562.65
- gender:degree	1	85.647	562.74
- religion:age	3	90.620	563.71

- religion:degree	1	88.643	565.74
+ gender:age	3	82.943	568.04
- country:age	9	106.955	568.05
- degree:country	3	94.978	568.07
- religion:country	3	98.028	571.12

Step: AIC=557.74

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + degree:age + country:age
```

	Df	Deviance	AIC
- degree:age	3	86.187	553.28
- gender:religion	1	86.222	557.32
<none>		84.643	557.74
- gender:degree	1	87.478	558.57
- religion:age	3	91.830	558.92
- religion:degree	1	89.634	560.73
+ gender:country	3	83.558	562.65
+ gender:age	3	83.965	563.06
- country:age	9	108.122	563.22
- degree:country	3	96.649	563.74
- religion:country	3	99.102	566.20

Step: AIC=553.28

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age
```

	Df	Deviance	AIC
- gender:religion	1	87.838	552.93
<none>		86.187	553.28
- gender:degree	1	88.958	554.05
- religion:age	3	93.428	554.52
- religion:degree	1	91.671	556.77
+ degree:age	3	84.643	557.74
+ gender:country	3	85.145	558.24
+ gender:age	3	85.421	558.52
- degree:country	3	97.463	558.56
- country:age	9	112.228	561.32
- religion:country	3	100.538	561.63

Step: AIC=552.93

```
cbind(yes, no) ~ gender + religion + degree + country + age +
  gender:degree + religion:degree + religion:country + religion:age +
  degree:country + country:age
```

	Df	Deviance	AIC
<none>		87.838	552.93
+ gender:religion	1	86.187	553.28
- gender:degree	1	90.686	553.78
- religion:age	3	94.842	553.94
- religion:degree	1	93.339	556.43
+ degree:age	3	86.222	557.32
+ gender:country	3	86.731	557.83
- degree:country	3	99.100	558.19
+ gender:age	3	87.174	558.27
- country:age	9	114.114	561.21
- religion:country	3	102.173	561.27

```
rm(full_model)

summary(model6)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
  age + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age, family = binomial(link = "cloglog"),
  data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.22334	0.11533	-1.937	0.052793 .
gender1	-0.10151	0.04461	-2.276	0.022865 *
religion1	0.28494	0.12148	2.346	0.018995 *
degree1	0.33068	0.16807	1.967	0.049126 *
countryNorway	-0.52275	0.19389	-2.696	0.007015 **
countrySweden	-0.34972	0.40255	-0.869	0.384979
countryUSA	-0.07423	0.15206	-0.488	0.625443
age2	-0.12601	0.14166	-0.890	0.373705
age3	-0.04896	0.16281	-0.301	0.763617
age4	-0.33287	0.19294	-1.725	0.084484 .
gender1:degree1	-0.16797	0.09938	-1.690	0.090993 .
religion1:degree1	-0.35498	0.14932	-2.377	0.017444 *

religion1:countryNorway	0.23029	0.17632	1.306	0.191509	
religion1:countrySweden	0.48064	0.39604	1.214	0.224895	
religion1:countryUSA	-0.34005	0.13242	-2.568	0.010231	*
religion1:age2	-0.24000	0.14715	-1.631	0.102898	
religion1:age3	-0.43077	0.16597	-2.596	0.009445	**
religion1:age4	-0.26948	0.19330	-1.394	0.163275	
degree1:countryNorway	0.13543	0.15723	0.861	0.389044	
degree1:countrySweden	-0.28609	0.16464	-1.738	0.082264	.
degree1:countryUSA	-0.17546	0.16008	-1.096	0.273032	
countryNorway:age2	0.34043	0.14181	2.401	0.016367	*
countrySweden:age2	0.27043	0.14544	1.859	0.062967	.
countryUSA:age2	0.20050	0.14571	1.376	0.168820	
countryNorway:age3	0.56026	0.15346	3.651	0.000261	***
countrySweden:age3	0.37774	0.15354	2.460	0.013886	*
countryUSA:age3	0.26853	0.16204	1.657	0.097482	.
countryNorway:age4	0.56008	0.16425	3.410	0.000650	***
countrySweden:age4	0.33267	0.16622	2.001	0.045350	*
countryUSA:age4	-0.07107	0.16695	-0.426	0.670315	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 87.838 on 88 degrees of freedom
 AIC: 552.93

Number of Fisher Scoring iterations: 5

37.

```
pchisq(deviance(model6), df.residual(model6), lower.tail = FALSE)
```

```
[1] 0.484825
```

38.

```

model6_df = data_frame()

WVS$model6_preds <- fitted(model6, type="response")
model6_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model6_preds,
                        pred_no = WVS$total * (1 - WVS$model6_preds))

sum(model6_df$pred_yes < 5)

```

[1] 34

```
sum(model6_df$pred_yes < 1)
```

[1] 6

```
sum(model6_df$pred_no < 5)
```

[1] 33

```
sum(model6_df$pred_no < 1)
```

[1] 7

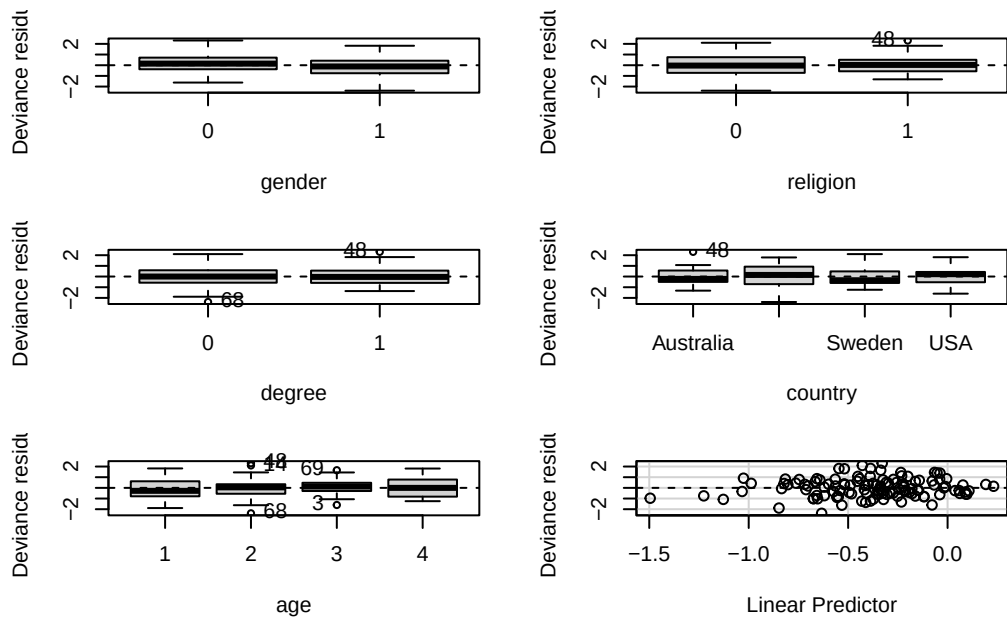
39.

```
vif(model6)
```

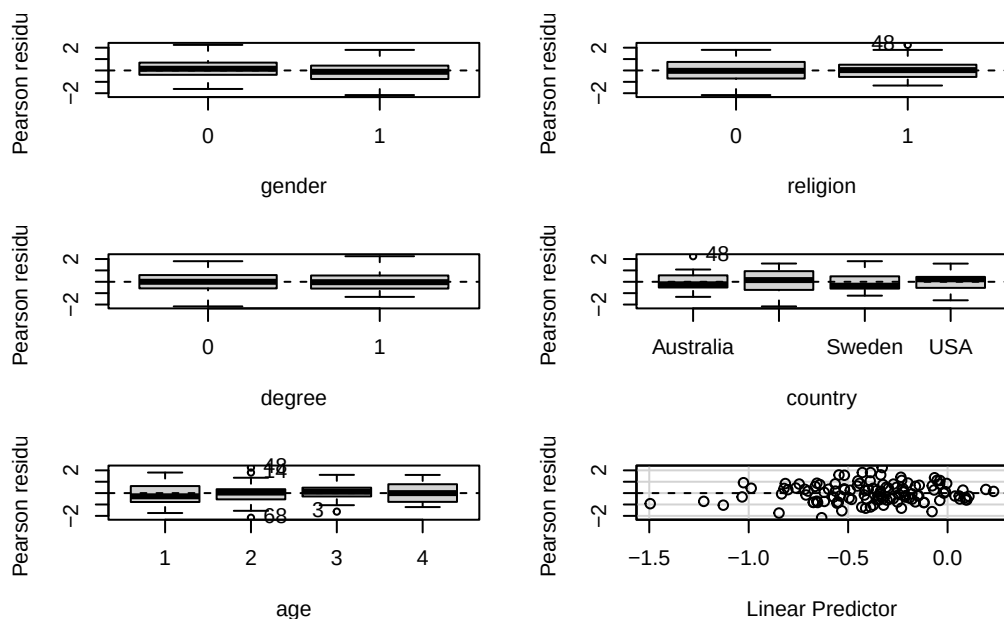
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.275898	1	1.129556
religion	4.741866	1	2.177583
degree	11.840512	1	3.441005
country	7519.963322	3	4.426250
age	750.708919	3	3.014708
gender:degree	2.038908	1	1.427903
religion:degree	8.195831	1	2.862836
religion:country	3612.978245	3	3.917216
religion:age	709.478993	3	2.986459
degree:country	15.341944	3	1.576329
country:age	1053.274422	9	1.472038

40.

```
residualPlots(model6, type = "deviance", smooth=FALSE)
```



```
residualPlots(model6, type = "pearson", smooth=FALSE)
```



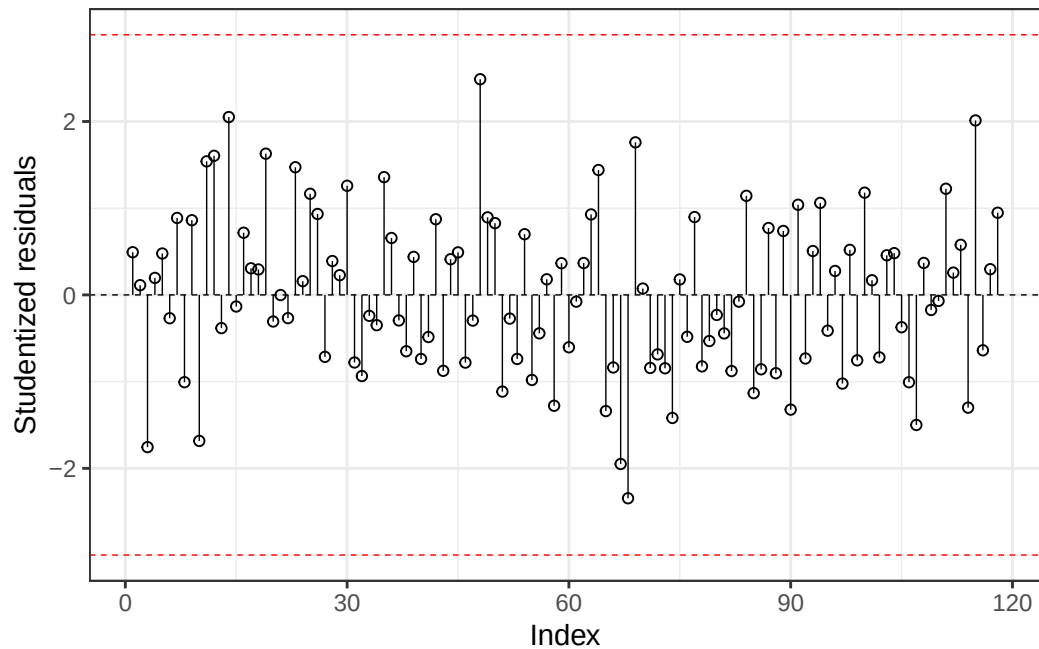
41.

```
p <- length(coef(model6))
n <- nobs(model6)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

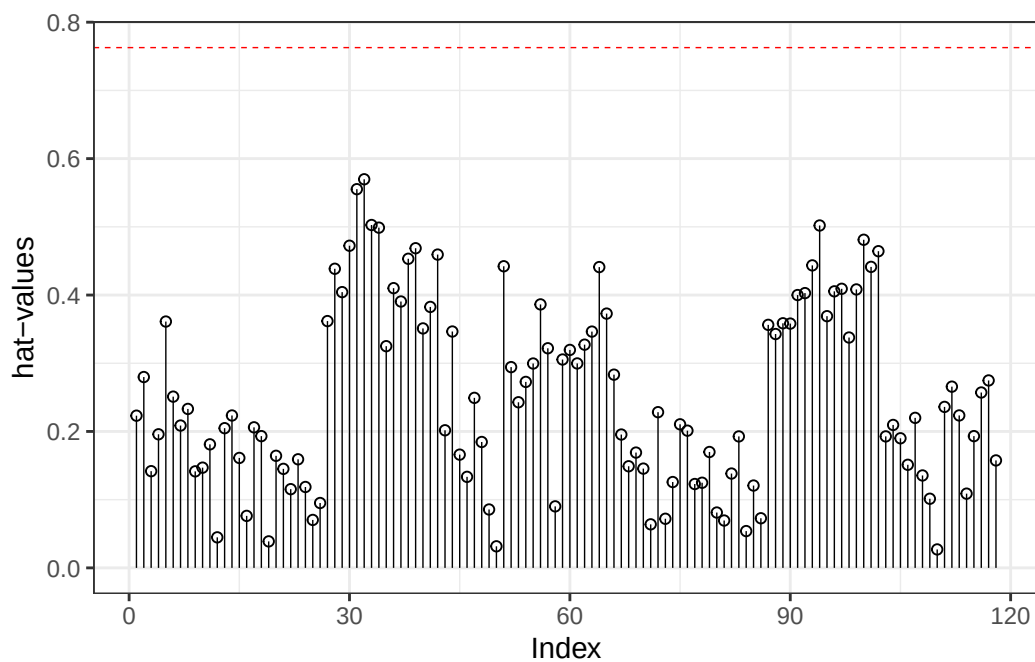
model6_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model6),
  hat = hatvalues(model6),
  cooks = cooks.distance(model6)
)

model6_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

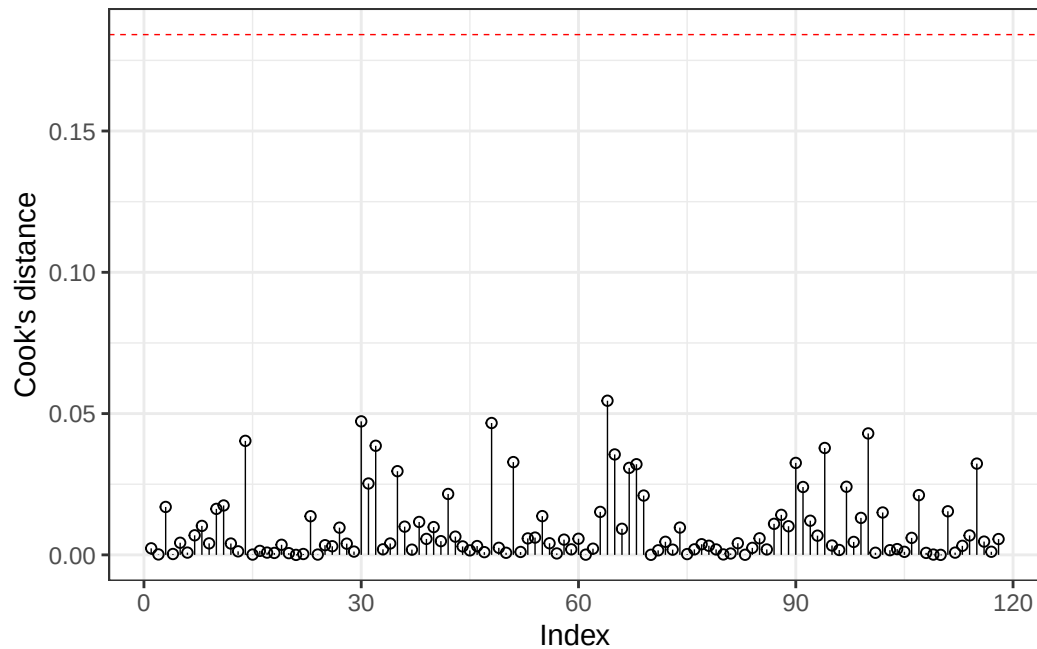
```
xlab("Index") +
ylab("Studentized residuals") +
theme_bw()
```



```
model6_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model6_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model6_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

42.

```
check_overdispersion(model6)
```

```
# Overdispersion test
```

```
dispersion ratio = 6487.022
Pearson's Chi-Squared = 570857.951
p-value = < 0.001
```

Model 7: loglog, all predictors

43.

```
model7 <- glm(cbind(no, yes) ~ gender + religion + degree + country + age, family = binomial)
summary(model7)
```

Call:

```
glm(formula = cbind(no, yes) ~ gender + religion + degree + country +
    age, family = binomial(link = "cloglog"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.72628	0.07185	-10.108	< 2e-16 ***
gender1	0.14432	0.03984	3.622	0.000292 ***
religion1	0.09566	0.05772	1.657	0.097456 .
degree1	0.12717	0.04979	2.554	0.010650 *
countryNorway	-0.10039	0.05719	-1.755	0.079194 .
countrySweden	-0.33364	0.06279	-5.314	1.08e-07 ***
countryUSA	0.24833	0.05045	4.922	8.57e-07 ***
age2	0.16861	0.05591	3.016	0.002563 **
age3	0.16998	0.06020	2.824	0.004746 **
age4	0.40628	0.05922	6.860	6.88e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.31 on 117 degrees of freedom
Residual deviance: 152.14 on 108 degrees of freedom
AIC: 577.24

Number of Fisher Scoring iterations: 5

44.

```
pchisq(deviance(model7), df.residual(model7), lower.tail = FALSE)
```

```
[1] 0.003317634
```

45.

```
model7_df = data_frame()

WVS$model7_preds <- fitted(model7, type="response")
model7_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model7_preds,
                        pred_no = WVS$total * (1 - WVS$model7_preds))

sum(model7_df$pred_yes < 5)
```

```
[1] 35
```

```
sum(model7_df$pred_yes < 1)
```

```
[1] 8
```

```
sum(model7_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model7_df$pred_no < 1)
```

```
[1] 6
```

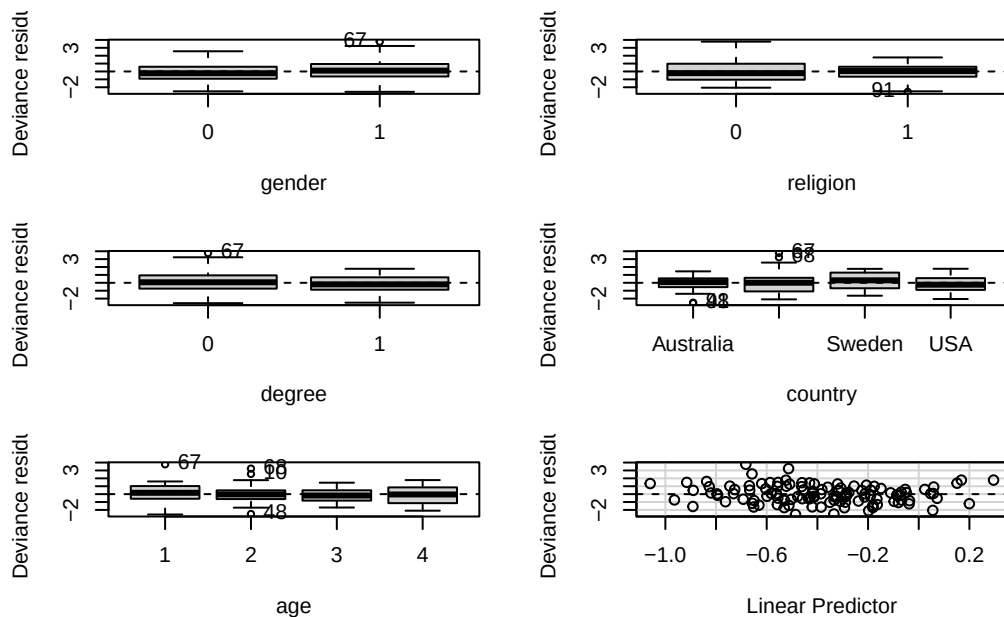
46.

```
vif(model7)
```

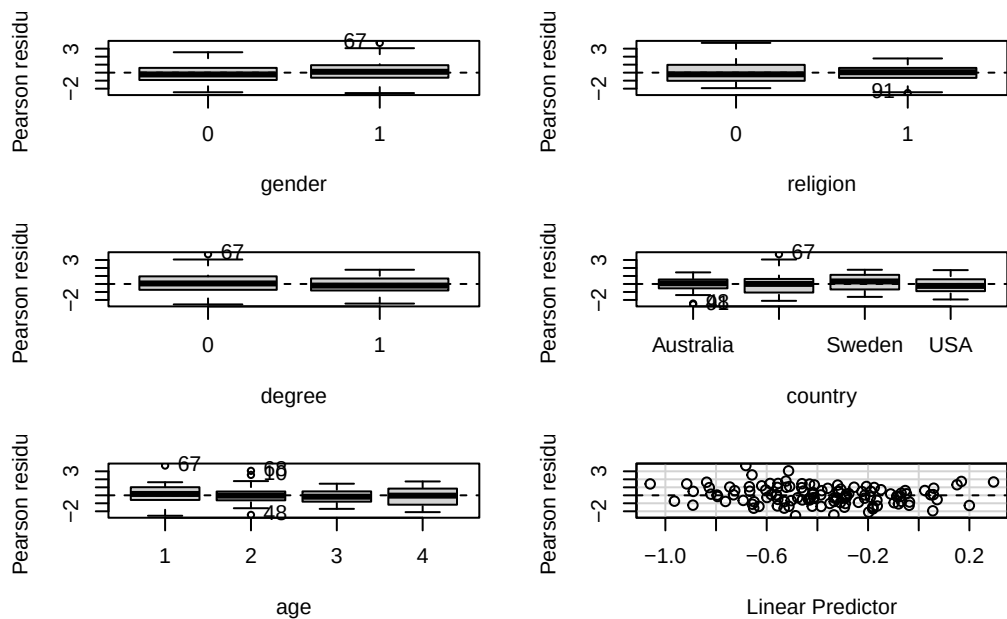
	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
gender	1.010502	1	1.005237
religion	1.059221	1	1.029184
degree	1.080618	1	1.039528
country	1.128528	3	1.020357
age	1.035811	3	1.005881

47.

```
residualPlots(model7, type = "deviance", smooth=FALSE)
```



```
residualPlots(model7, type = "pearson", smooth=FALSE)
```

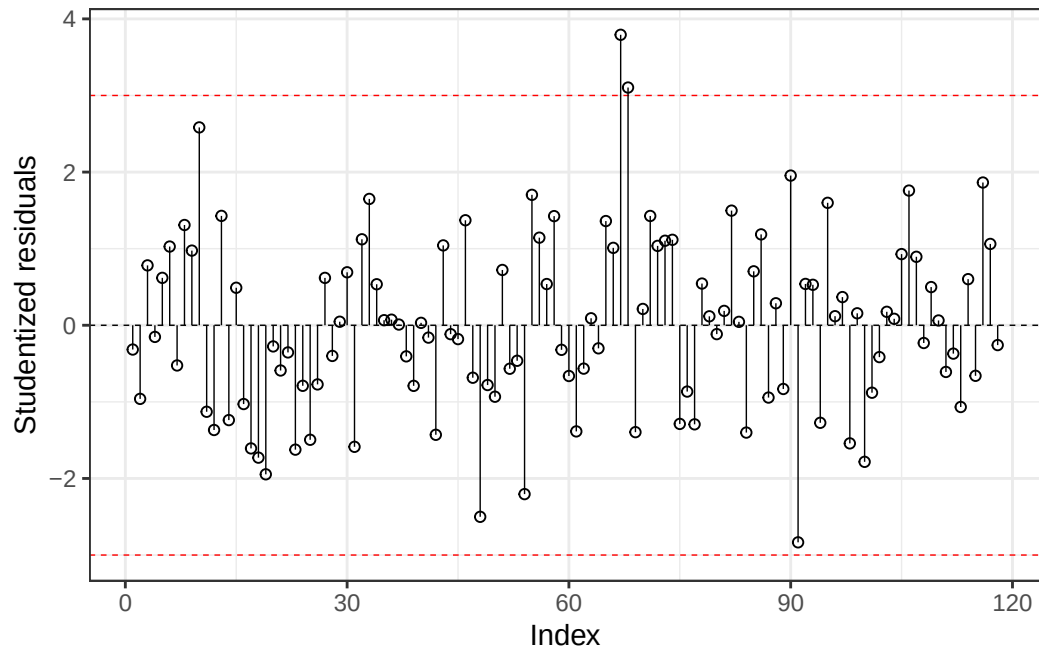
48.

```
p <- length(coef(model7))
n <- nobs(model7)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

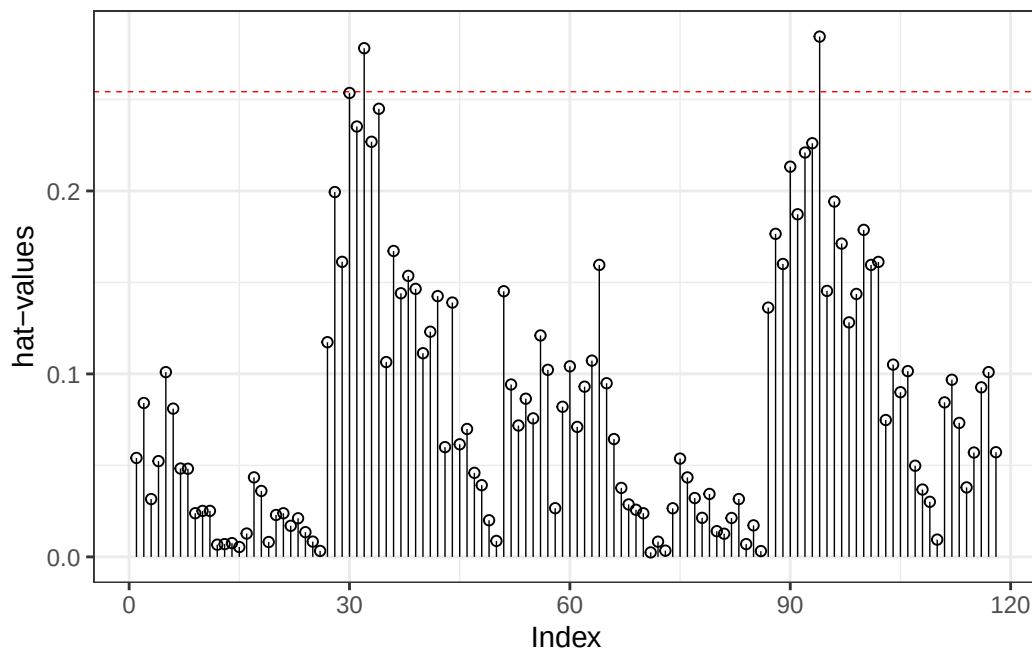
model7_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model7),
  hat = hatvalues(model7),
  cooks = cooks.distance(model7)
)

model7_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

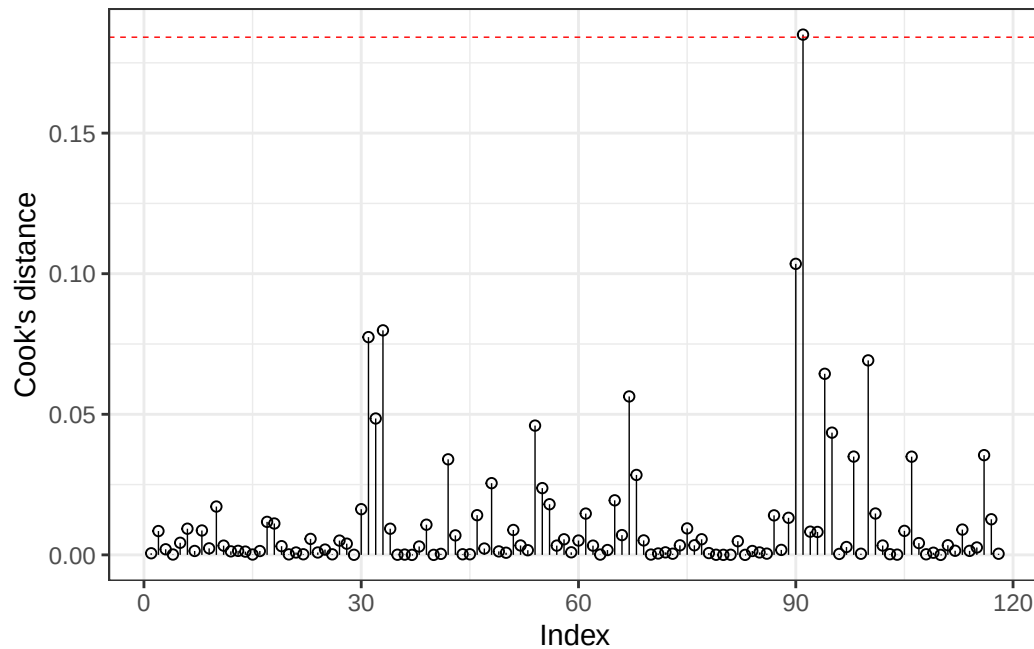
```
xlab("Index") +
ylab("Studentized residuals") +
theme_bw()
```



```
model7_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model7_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model7_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

	obs	studentized	hat	cooks
	32	1.122383	0.27799643	0.04850448
	67	3.792334	0.03770319	0.05634848
	68	3.103774	0.02866908	0.02843327
	91	-2.833919	0.18725015	0.18502911
	94	-1.273216	0.28434220	0.06440802

49.

```
check_overdispersion(model7)
```

```
# Overdispersion test
```

```
dispersion ratio = 5206.738
```

```
Pearson's Chi-Squared = 562327.755
p-value = < 0.001
```

Model 8: loglog, stepwise selected predictors

50.

```
full_model <- glm(cbind(no, yes) ~ (gender + religion + degree + country + age)^2, family = l
model8 <- step(full_model, direction = "both")
```

Start: AIC=564.38

```
cbind(no, yes) ~ (gender + religion + degree + country + age)^2
```

	Df	Deviance	AIC
- gender:age	3	79.738	558.83
- gender:country	3	80.535	559.63
- degree:age	3	81.056	560.15
- gender:degree	1	81.074	564.17
<none>		79.281	564.38
- gender:religion	1	81.366	564.46
- religion:age	3	87.327	566.42
- country:age	9	99.931	567.03
- religion:degree	1	84.116	567.21
- degree:country	3	91.450	570.54
- religion:country	3	95.395	574.49

Step: AIC=558.83

```
cbind(no, yes) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + gender:country + religion:degree +
  religion:country + religion:age + degree:country + degree:age +
  country:age
```

	Df	Deviance	AIC
- gender:country	3	80.962	554.06
- degree:age	3	81.540	554.63
- gender:degree	1	81.388	558.48
- gender:religion	1	81.654	558.75
<none>		79.738	558.83
- religion:age	3	87.624	560.72

- country:age	9	100.288	561.38
- religion:degree	1	84.573	561.67
+ gender:age	3	79.281	564.38
- degree:country	3	91.983	565.08
- religion:country	3	95.875	568.97

Step: AIC=554.06

```
cbind(no, yes) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + degree:age + country:age
```

	Df	Deviance	AIC
- degree:age	3	82.703	549.80
- gender:religion	1	82.774	553.87
<none>		80.962	554.06
- gender:degree	1	83.335	554.43
- religion:age	3	88.884	555.98
- country:age	9	101.515	556.61
- religion:degree	1	85.770	556.86
+ gender:country	3	79.738	558.83
+ gender:age	3	80.535	559.63
- degree:country	3	93.427	560.52
- religion:country	3	97.095	564.19

Step: AIC=549.8

```
cbind(no, yes) ~ gender + religion + degree + country + age +
  gender:religion + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age
```

	Df	Deviance	AIC
- gender:religion	1	84.541	549.64
<none>		82.703	549.80
- gender:degree	1	85.145	550.24
- religion:age	3	90.289	551.38
- religion:degree	1	87.636	552.73
+ degree:age	3	80.962	554.06
+ gender:country	3	81.540	554.63
- country:age	9	106.017	555.11
+ gender:age	3	82.252	555.35
- degree:country	3	94.645	555.74
- religion:country	3	98.711	559.81

Step: AIC=549.64

```
cbind(no, yes) ~ gender + religion + degree + country + age +
  gender:degree + religion:degree + religion:country + religion:age +
  degree:country + country:age
```

	Df	Deviance	AIC
<none>		84.541	549.64
+ gender:religion	1	82.703	549.80
- gender:degree	1	86.850	549.94
- religion:age	3	91.831	550.93
- religion:degree	1	89.417	552.51
+ degree:age	3	82.774	553.87
+ gender:country	3	83.493	554.59
- country:age	9	108.011	555.11
+ gender:age	3	84.231	555.33
- degree:country	3	96.423	555.52
- religion:country	3	100.118	559.21

```
rm(full_model)

summary(model8)
```

Call:

```
glm(formula = cbind(no, yes) ~ gender + religion + degree + country +
  age + gender:degree + religion:degree + religion:country +
  religion:age + degree:country + country:age, family = binomial(link = "cloglog"),
  data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.53610	0.12127	-4.421	9.84e-06 ***
gender1	0.11112	0.04527	2.455	0.014095 *
religion1	-0.30812	0.12862	-2.396	0.016591 *
degree1	-0.32880	0.18125	-1.814	0.069667 .
countryNorway	0.58619	0.18318	3.200	0.001374 **
countrySweden	0.38640	0.38529	1.003	0.315906
countryUSA	0.11857	0.15359	0.772	0.440144
age2	0.15097	0.14461	1.044	0.296482
age3	0.06285	0.17029	0.369	0.712097
age4	0.36636	0.17562	2.086	0.036971 *
gender1:degree1	0.14725	0.09705	1.517	0.129225
religion1:degree1	0.33276	0.15260	2.181	0.029213 *

```

religion1:countryNorway -0.25626    0.16806   -1.525  0.127305
religion1:countrySweden -0.56415    0.37633   -1.499  0.133846
religion1:countryUSA     0.33517    0.13126    2.554  0.010663 *
religion1:age2           0.25694    0.14853    1.730  0.083651 .
religion1:age3           0.45804    0.17263    2.653  0.007971 **
religion1:age4           0.26028    0.17456    1.491  0.135949
degree1:countryNorway   -0.11967    0.16545   -0.723  0.469482
degree1:countrySweden   0.36184    0.17484    2.070  0.038492 *
degree1:countryUSA      0.15952    0.15606    1.022  0.306706
countryNorway:age2      -0.36931    0.15120   -2.443  0.014584 *
countrySweden:age2      -0.27375    0.17458   -1.568  0.116880
countryUSA:age2         -0.25110    0.14594   -1.721  0.085331 .
countryNorway:age3      -0.60921    0.16512   -3.689  0.000225 ***
countrySweden:age3      -0.38713    0.18035   -2.147  0.031833 *
countryUSA:age3         -0.33500    0.15882   -2.109  0.034914 *
countryNorway:age4      -0.58127    0.16983   -3.423  0.000620 ***
countrySweden:age4      -0.29395    0.18841   -1.560  0.118720
countryUSA:age4         -0.06531    0.15068   -0.433  0.664680
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 319.307  on 117  degrees of freedom
Residual deviance: 84.541  on 88  degrees of freedom
AIC: 549.64

```

Number of Fisher Scoring iterations: 5

51.

```
pchisq(deviance(model8), df.residual(model8), lower.tail = FALSE)
```

```
[1] 0.5846489
```

52.


```

model8_df = data_frame()

WVS$model8_preds <- fitted(model2, type="response")
model8_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model8_preds,
                        pred_no = WVS$total * (1 - WVS$model8_preds))

sum(model8_df$pred_yes < 5)

```

[1] 36

```
sum(model8_df$pred_yes < 1)
```

[1] 6

```
sum(model8_df$pred_no < 5)
```

[1] 33

```
sum(model8_df$pred_no < 1)
```

[1] 7

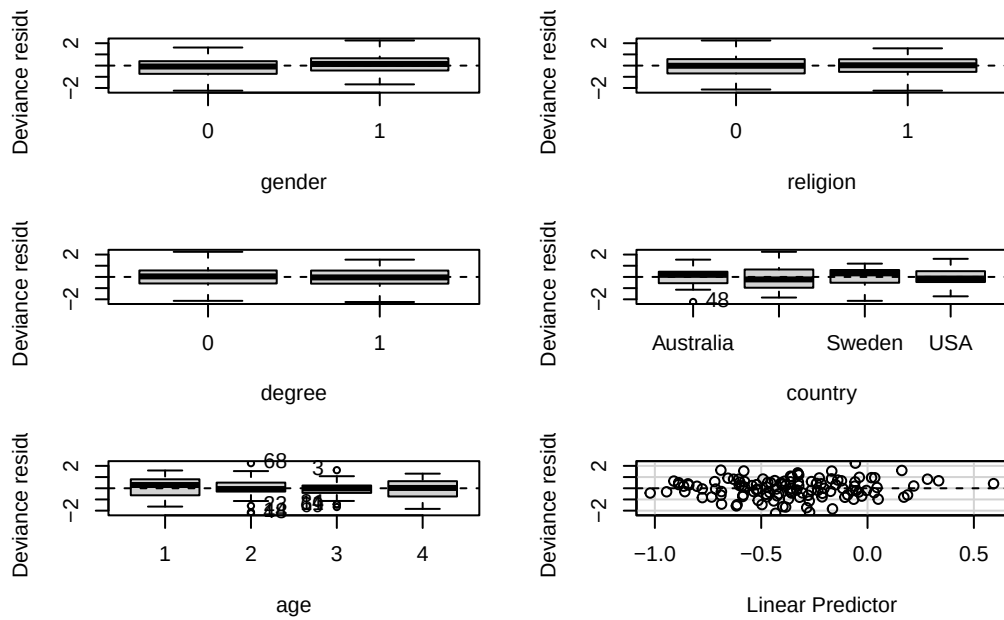
53.

```
vif(model8)
```

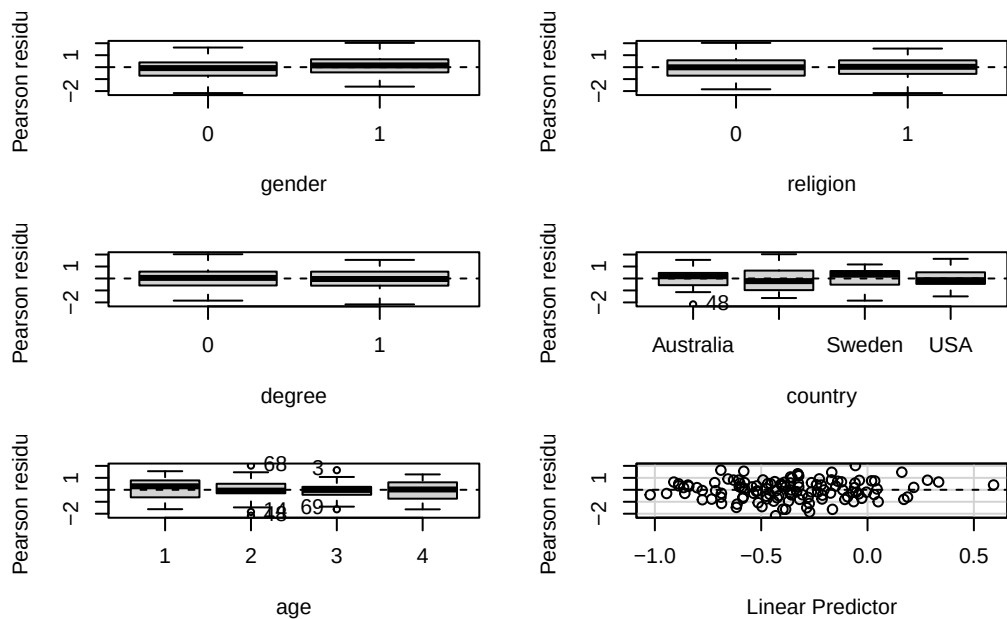
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.298656	1	1.139586
religion	5.302542	1	2.302725
degree	14.214396	1	3.770198
country	5171.212860	3	4.158456
age	635.608913	3	2.932232
gender:degree	2.380312	1	1.542826
religion:degree	9.129004	1	3.021424
religion:country	2676.985485	3	3.726270
religion:age	642.817758	3	2.937749
degree:country	18.726153	3	1.629577
country:age	1357.236792	9	1.492920

54.

```
residualPlots(model8, type = "deviance", smooth=FALSE)
```



```
residualPlots(model8, type = "pearson", smooth=FALSE)
```



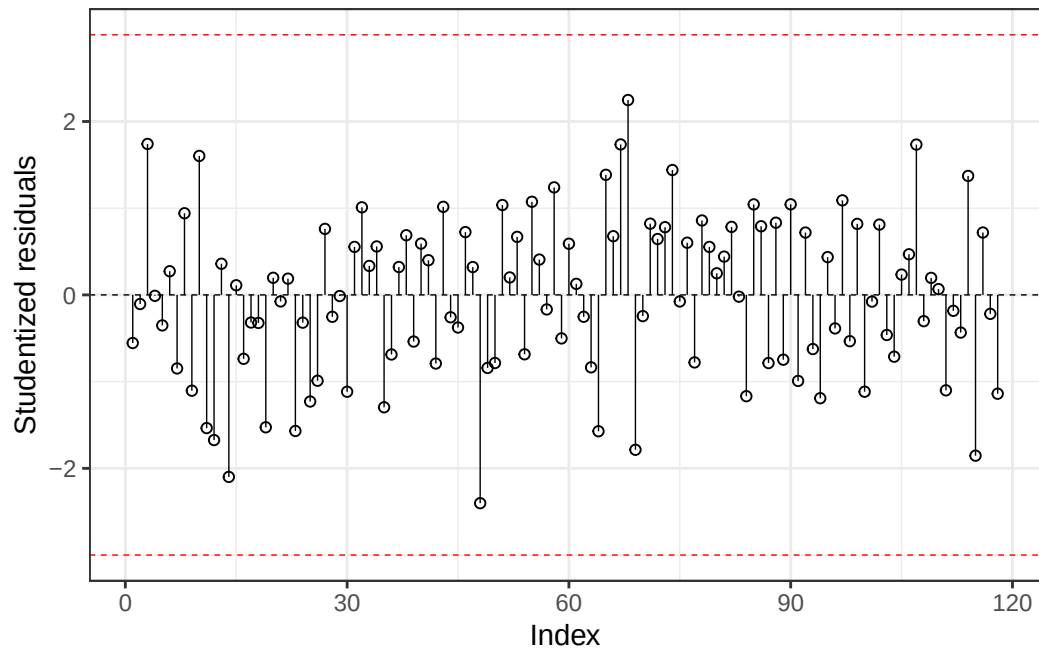
55.

```
p <- length(coef(model8))
n <- nobs(model8)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

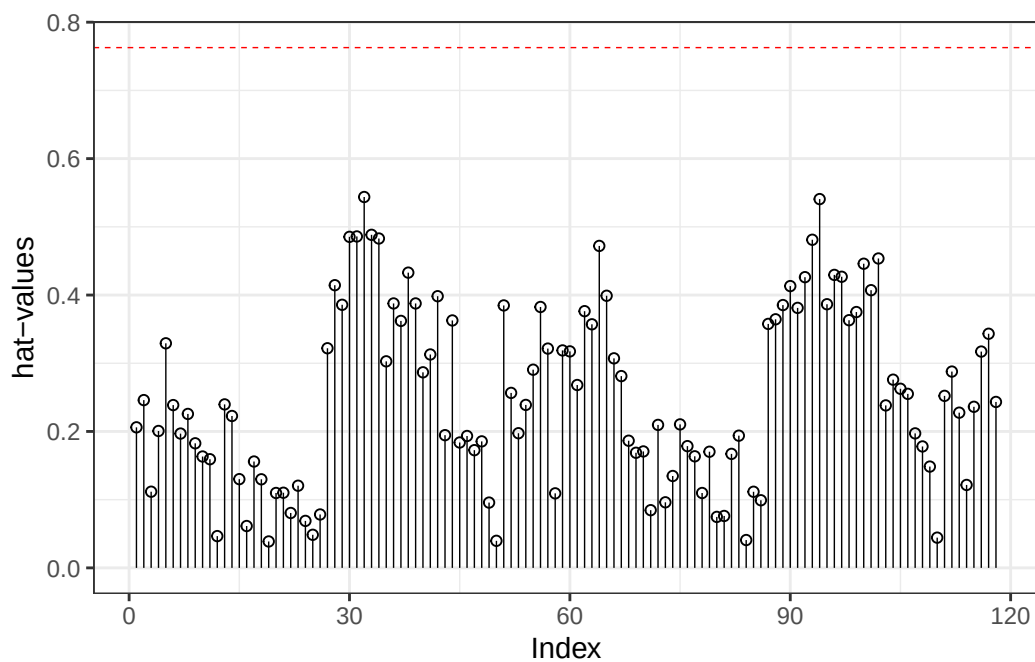
model8_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model8),
  hat = hatvalues(model8),
  cooks = cooks.distance(model8)
)

model8_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

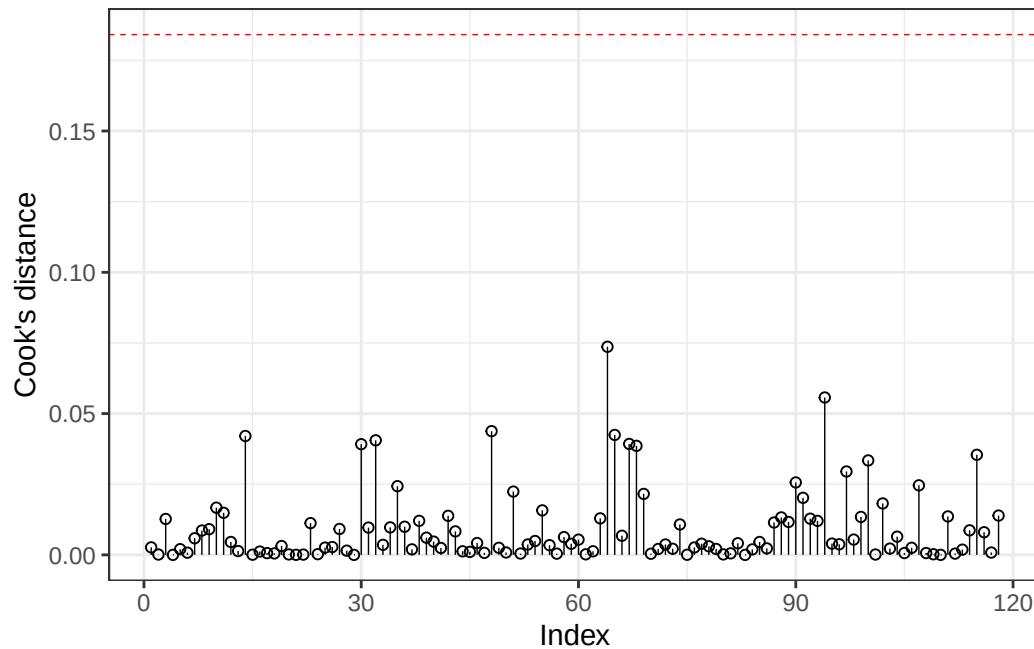
```
xlab("Index") +
ylab("Studentized residuals") +
theme_bw()
```



```
model8_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model8_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model8_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

56.

```
check_overdispersion(model8)
```

```
# Overdispersion test
```

```
dispersion ratio = 6502.067
Pearson's Chi-Squared = 572181.903
p-value = < 0.001
```

Model 2, 4, 6, 8 AIC comparisons

57.

```
AIC(model2)
```

```
[1] 551.4403
```

```
AIC(model4)
```

```
[1] 551.1816
```

```
AIC(model6)
```

```
[1] 552.9323
```

```
AIC(model8)
```

```
[1] 549.6353
```

Models 9-10: logit, manually reduced variables

58.

```
model2_lrt <- Anova(model2)
model2_lrt_p <- model2_lrt[, "Pr(>Chisq)"]

cbind(model2_lrt, p.adj = p.adjust(model2_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR	Chisq	Df	Pr(>Chisq)	p.adj
religion	1.242416	1	0.26500590	0.26500590	
gender:degree	2.626733	1	0.10507771	0.11558548	
religion:age	7.135573	3	0.06769986	0.08274427	

59.

```
model9 <- update(model2, . ~ . - gender:degree)
summary(model9)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age + religion:degree + religion:country + religion:age +
    degree:country + country:age, family = binomial(link = "logit"),
    data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.24183	0.16381	1.476	0.139854
gender1	-0.19582	0.05620	-3.484	0.000493 ***
religion1	0.40909	0.17444	2.345	0.019017 *
degree1	0.32939	0.22992	1.433	0.151955
countryNorway	-0.77436	0.26538	-2.918	0.003524 **
countrySweden	-0.51583	0.55190	-0.935	0.349973
countryUSA	-0.13215	0.21267	-0.621	0.534362
age2	-0.18760	0.19913	-0.942	0.346142
age3	-0.06508	0.23140	-0.281	0.778533
age4	-0.47867	0.25667	-1.865	0.062193 .
religion1:degree1	-0.45138	0.21035	-2.146	0.031883 *
religion1:countryNorway	0.35020	0.24186	1.448	0.147623
religion1:countrySweden	0.73809	0.54102	1.364	0.172483
religion1:countryUSA	-0.46894	0.18367	-2.553	0.010674 *
religion1:age2	-0.35096	0.20628	-1.701	0.088870 .
religion1:age3	-0.63019	0.23569	-2.674	0.007499 **
religion1:age4	-0.37322	0.25670	-1.454	0.145976
degree1:countryNorway	0.18384	0.22477	0.818	0.413416
degree1:countrySweden	-0.45443	0.23647	-1.922	0.054642 .
degree1:countryUSA	-0.24031	0.22106	-1.087	0.277005
countryNorway:age2	0.48666	0.20405	2.385	0.017082 *
countrySweden:age2	0.38184	0.22380	1.706	0.087978 .
countryUSA:age2	0.31671	0.20256	1.564	0.117932
countryNorway:age3	0.80528	0.22157	3.634	0.000279 ***
countrySweden:age3	0.52929	0.23319	2.270	0.023224 *
countryUSA:age3	0.41701	0.22312	1.869	0.061625 .
countryNorway:age4	0.78680	0.23234	3.386	0.000708 ***
countrySweden:age4	0.42174	0.24739	1.705	0.088236 .
countryUSA:age4	-0.01385	0.22083	-0.063	0.950009

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
Residual deviance: 88.972 on 89 degrees of freedom
AIC: 552.07

Number of Fisher Scoring iterations: 4

60.

```
anova(model2, model9)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
gender:degree + religion:degree + religion:country + religion:age +
degree:country + country:age

Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +
religion:degree + religion:country + religion:age + degree:country +
country:age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	88	86.346			
2	89	88.972	-1	-2.6267	0.1051

61.

```
model9_lrt <- Anova(model9)
model9_lrt_p <- model9_lrt[, "Pr(>Chisq)"]

cbind(model9_lrt, p.adj = p.adjust(model9_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR	Chisq	Df	Pr(>Chisq)	p.adj
religion	1.261986	1	0.26127549	0.26127549	
religion:age	7.414381	3	0.05979966	0.06644407	

62.

```
model10 <- update(model9, . ~ . - religion:age)
summary(model10)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age + religion:degree + religion:country + degree:country +
    country:age, family = binomial(link = "logit"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.464381	0.135927	3.416	0.000635	***
gender1	-0.194751	0.056148	-3.469	0.000523	***
religion1	0.107397	0.120573	0.891	0.373079	
degree1	0.357344	0.229829	1.555	0.119988	
countryNorway	-0.722736	0.262813	-2.750	0.005960	**
countrySweden	-0.377472	0.548017	-0.689	0.490952	
countryUSA	-0.111879	0.213022	-0.525	0.599445	
age2	-0.449527	0.125118	-3.593	0.000327	***
age3	-0.560099	0.137342	-4.078	4.54e-05	***
age4	-0.765665	0.139838	-5.475	4.37e-08	***
religion1:degree1	-0.472419	0.209384	-2.256	0.024056	*
religion1:countryNorway	0.342622	0.240680	1.424	0.154574	
religion1:countrySweden	0.671931	0.538456	1.248	0.212073	
religion1:countryUSA	-0.498727	0.182782	-2.729	0.006362	**
degree1:countryNorway	0.182686	0.224336	0.814	0.415450	
degree1:countrySweden	-0.460818	0.236069	-1.952	0.050933	.
degree1:countryUSA	-0.238187	0.220753	-1.079	0.280600	
countryNorway:age2	0.433782	0.201353	2.154	0.031214	*
countrySweden:age2	0.298672	0.218263	1.368	0.171185	
countryUSA:age2	0.316444	0.202801	1.560	0.118674	
countryNorway:age3	0.731798	0.219398	3.335	0.000852	***
countrySweden:age3	0.406447	0.228405	1.779	0.075158	.
countryUSA:age3	0.427718	0.222711	1.921	0.054794	.
countryNorway:age4	0.737769	0.230792	3.197	0.001390	**
countrySweden:age4	0.341820	0.243559	1.403	0.160487	
countryUSA:age4	-0.006884	0.220550	-0.031	0.975102	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
Residual deviance: 96.387 on 92 degrees of freedom
AIC: 553.48

Number of Fisher Scoring iterations: 4

63.

```
anova(model9, model10)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
religion:degree + religion:country + religion:age + degree:country +
country:age

Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +
religion:degree + religion:country + degree:country + country:age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	89	88.972			
2	92	96.387	-3	-7.4144	0.0598 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

64.

```
model10_lrt <- Anova(model10)
model9_lrt_10 <- model10_lrt[, "Pr(>Chisq)"]

cbind(model10_lrt, p.adj = p.adjust(model9_lrt_10, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR	Chisq	Df	Pr(>Chisq)	p.adj
religion	1.261986	1	0.2612755	0.2612755	

65.

```
pchisq(deviance(model10), df.residual(model10), lower.tail = FALSE)
```

```
[1] 0.3566673
```

66.

```
model10_df = data_frame()

WVS$model10_preds <- fitted(model10, type="response")
model10_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model10_preds,
                        pred_no = WVS$total * (1 - WVS$model10_preds))

sum(model10_df$pred_yes < 5)
```

```
[1] 35
```

```
sum(model10_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model10_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model10_df$pred_no < 1)
```

```
[1] 6
```

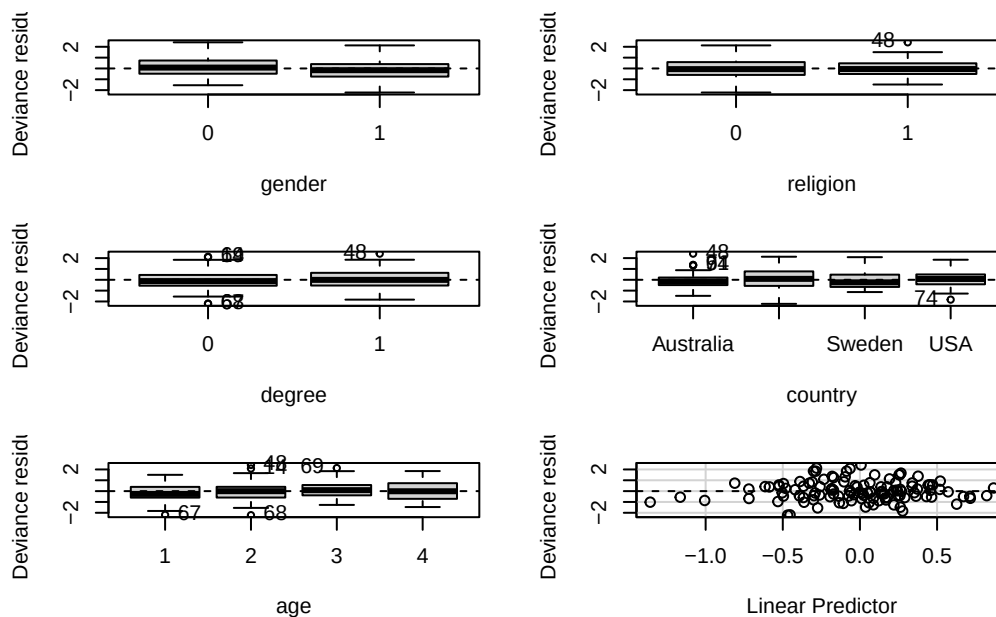
67.

```
vif(model10)
```

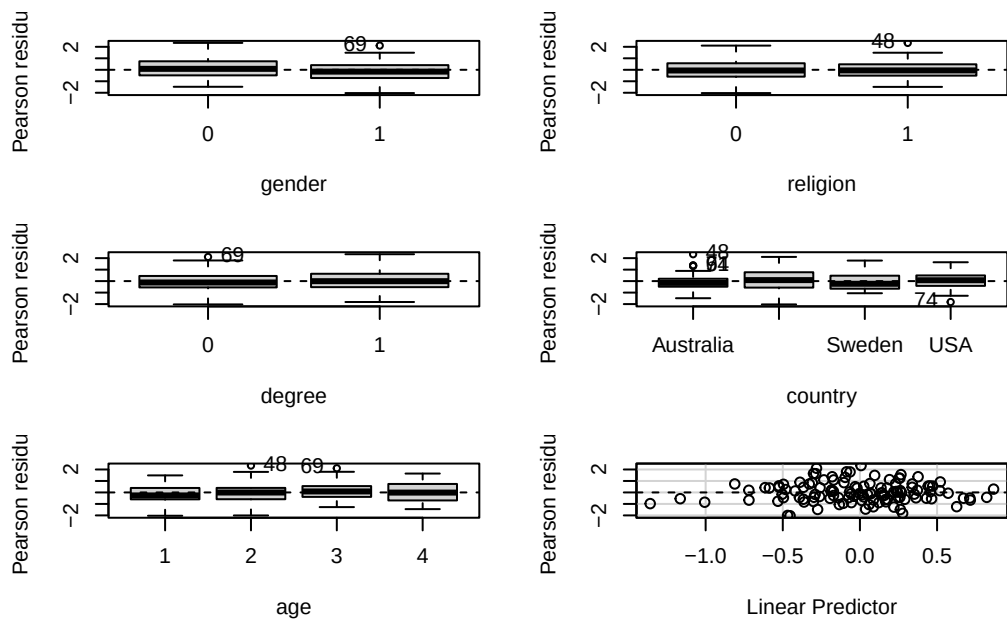
	GVIF	Df	GVIF ^{1/(2*Df)}
gender	1.017085	1	1.008506
religion	2.374020	1	1.540785
degree	11.371798	1	3.372210
country	6054.412060	3	4.269188
age	23.159735	3	1.688323
religion:degree	8.410858	1	2.900148
religion:country	3007.774991	3	3.799335
degree:country	16.558080	3	1.596498
country:age	963.443875	9	1.464766

68.

```
residualPlots(model10, type = "deviance", smooth=FALSE)
```



```
residualPlots(model10, type = "pearson", smooth=FALSE)
```



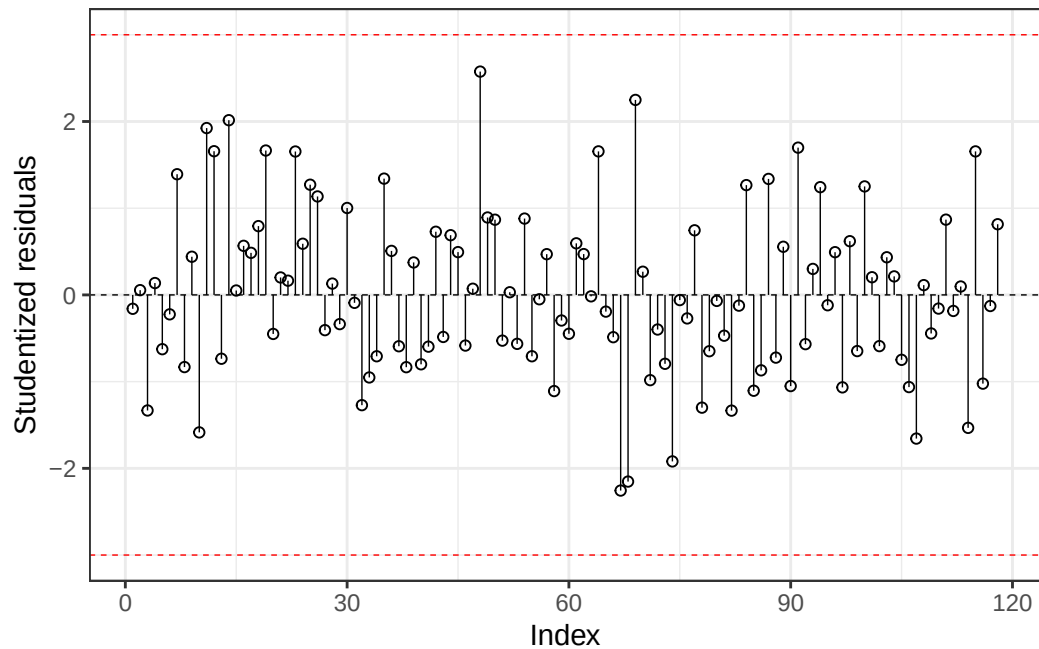
69.

```
p <- length(coef(model10))
n <- nobs(model10)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

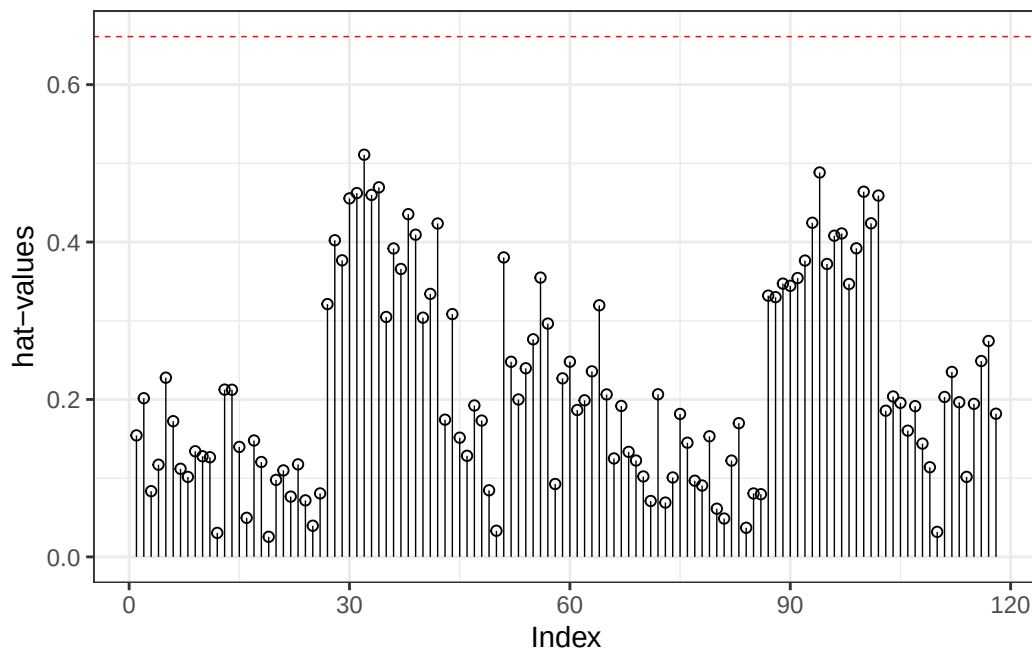
model10_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model10),
  hat = hatvalues(model10),
  cooks = cooks.distance(model10)
)

model10_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
```

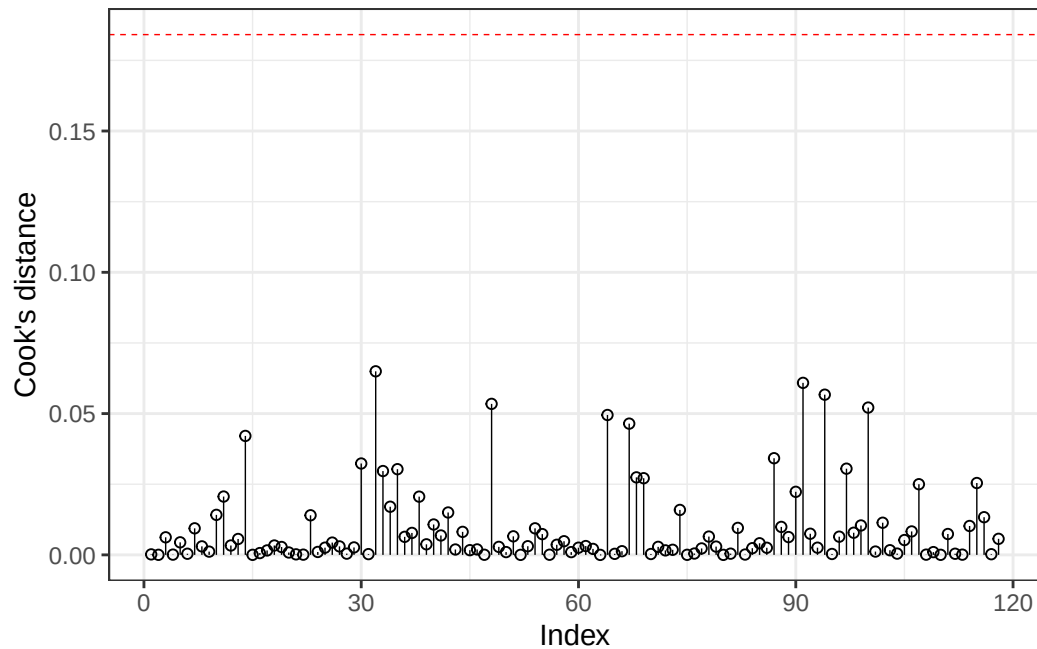
```
xlab("Index") +
ylab("Studentized residuals") +
theme_bw()
```



```
model10_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model10_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```

```
model10_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

70.

```
check_overdispersion(model10)
```

```
# Overdispersion test
```

```
dispersion ratio = 6194.860
Pearson's Chi-Squared = 569927.102
p-value = < 0.001
```

Models 11-12: probit, manually reduced variables

71.

```
model4_lrt <- Anova(model4)
model4_lrt_p <- model4_lrt[, "Pr(>Chisq)"]

cbind(model4_lrt, p.adj = p.adjust(model4_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR Chisq	Df	Pr(>Chisq)	p.adj
religion	1.304316	1	0.25342635	0.2534264
gender:degree	2.608607	1	0.10628506	0.1169136
religion:age	7.136588	3	0.06766936	0.0827070

72.

```
model11 <- update(model4, . ~ . - gender:degree)
summary(model11)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age + religion:degree + religion:country + religion:age +
    degree:country + country:age, family = binomial(link = "probit"),
    data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.150459	0.102249	1.471	0.141158
gender1	-0.122055	0.034907	-3.497	0.000471 ***
religion1	0.254912	0.108660	2.346	0.018978 *
degree1	0.205862	0.143032	1.439	0.150073
countryNorway	-0.484628	0.164799	-2.941	0.003274 **
countrySweden	-0.325473	0.344339	-0.945	0.344550
countryUSA	-0.082923	0.132649	-0.625	0.531888
age2	-0.116562	0.124346	-0.937	0.348552
age3	-0.039899	0.144413	-0.276	0.782332
age4	-0.299777	0.159541	-1.879	0.060246 .

religion1:degree1	-0.282744	0.130755	-2.162	0.030588	*
religion1:countryNorway	0.220495	0.150448	1.466	0.142760	
religion1:countrySweden	0.462008	0.337756	1.368	0.171351	
religion1:countryUSA	-0.291550	0.114533	-2.546	0.010911	*
religion1:age2	-0.218723	0.128680	-1.700	0.089180	.
religion1:age3	-0.393460	0.146922	-2.678	0.007406	**
religion1:age4	-0.231119	0.159370	-1.450	0.147002	
degree1:countryNorway	0.115960	0.140090	0.828	0.407810	
degree1:countrySweden	-0.281149	0.147138	-1.911	0.056031	.
degree1:countryUSA	-0.148402	0.136997	-1.083	0.278697	
countryNorway:age2	0.303072	0.127194	2.383	0.017184	*
countrySweden:age2	0.239151	0.138222	1.730	0.083596	.
countryUSA:age2	0.197242	0.126240	1.562	0.118184	
countryNorway:age3	0.502548	0.138040	3.641	0.000272	***
countrySweden:age3	0.331190	0.144101	2.298	0.021544	*
countryUSA:age3	0.259756	0.138904	1.870	0.061478	.
countryNorway:age4	0.490908	0.144823	3.390	0.000700	***
countrySweden:age4	0.265622	0.153025	1.736	0.082598	.
countryUSA:age4	-0.004709	0.136254	-0.035	0.972431	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 88.696 on 89 degrees of freedom
 AIC: 551.79

Number of Fisher Scoring iterations: 4

73.

```
anova(model4, model11)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
 gender:degree + religion:degree + religion:country + religion:age +
 degree:country + country:age
 Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +

```

religion:degree + religion:country + religion:age + degree:country +
country:age
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      88      86.087
2      89      88.696 -1   -2.6086  0.1063

```

74.

```

model11_lrt <- Anova(model11)
model11_lrt_p <- model11_lrt[, "Pr(>Chisq)"]

cbind(model11_lrt, p.adj = p.adjust(model11_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

	LR	Chisq	Df	Pr(>Chisq)	p.adj
religion	1.320768	1	0.25045418	0.25045418	
religion:age	7.411073	3	0.05988792	0.06654214	

75.

```

model12 <- update(model11, . ~ . - religion:age)
summary(model12)

```

Call:

```

glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
  age + religion:degree + religion:country + degree:country +
  country:age, family = binomial(link = "probit"), data = WVS)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.2890352	0.0845035	3.420	0.000625	***
gender1	-0.1214111	0.0348841	-3.480	0.000501	***
religion1	0.0677745	0.0751763	0.902	0.367301	
degree1	0.2219674	0.1428581	1.554	0.120241	
countryNorway	-0.4518985	0.1636825	-2.761	0.005766	**
countrySweden	-0.2377640	0.3422354	-0.695	0.487219	
countryUSA	-0.0694431	0.1326003	-0.524	0.600486	

age2	-0.2802523	0.0778590	-3.599	0.000319	***
age3	-0.3495852	0.0855273	-4.087	4.36e-05	***
age4	-0.4779997	0.0868915	-5.501	3.77e-08	***
religion1:degree1	-0.2950340	0.1301848	-2.266	0.023435	*
religion1:countryNorway	0.2144885	0.1499846	1.430	0.152697	
religion1:countrySweden	0.4187560	0.3364230	1.245	0.213231	
religion1:countryUSA	-0.3116146	0.1138256	-2.738	0.006188	**
degree1:countryNorway	0.1157558	0.1399260	0.827	0.408087	
degree1:countrySweden	-0.2846389	0.1469849	-1.937	0.052804	.
degree1:countryUSA	-0.1467184	0.1368775	-1.072	0.283767	
countryNorway:age2	0.2704907	0.1257048	2.152	0.031414	*
countrySweden:age2	0.1875597	0.1348648	1.391	0.164309	
countryUSA:age2	0.1973186	0.1263266	1.562	0.118295	
countryNorway:age3	0.4571085	0.1368643	3.340	0.000838	***
countrySweden:age3	0.2547865	0.1412196	1.804	0.071202	.
countryUSA:age3	0.2666935	0.1387453	1.922	0.054583	.
countryNorway:age4	0.4610981	0.1440287	3.201	0.001367	**
countrySweden:age4	0.2164270	0.1507652	1.436	0.151138	
countryUSA:age4	-0.0001148	0.1361032	-0.001	0.999327	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 96.107 on 92 degrees of freedom
 AIC: 553.2

Number of Fisher Scoring iterations: 4

76.

```
anova(model11, model12)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
 religion:degree + religion:country + religion:age + degree:country +
 country:age
 Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +

```

    religion:degree + religion:country + degree:country + country:age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         89      88.696
2         92      96.107 -3  -7.4111  0.05989 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

77.

```

model12_lrt <- Anova(model12)
model12_lrt_p <- model12_lrt[, "Pr(>Chisq)"]

cbind(model12_lrt, p.adj = p.adjust(model12_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

```

      LR Chisq Df Pr(>Chisq)    p.adj
religion 1.320768  1  0.2504542 0.2504542

```

78.

```

pchisq(deviance(model12), df.residual(model12), lower.tail = FALSE)

```

```

[1] 0.3641615

```

79.

```

model12_df = data_frame()

WVS$model12_preds <- fitted(model12, type="response")
model12_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model12_preds,
                        pred_no = WVS$total * (1 - WVS$model12_preds))

sum(model12_df$pred_yes < 5)

```

```
[1] 35
```

```
sum(model12_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model12_df$pred_no < 5)
```

```
[1] 33
```

```
sum(model12_df$pred_no < 1)
```

```
[1] 6
```

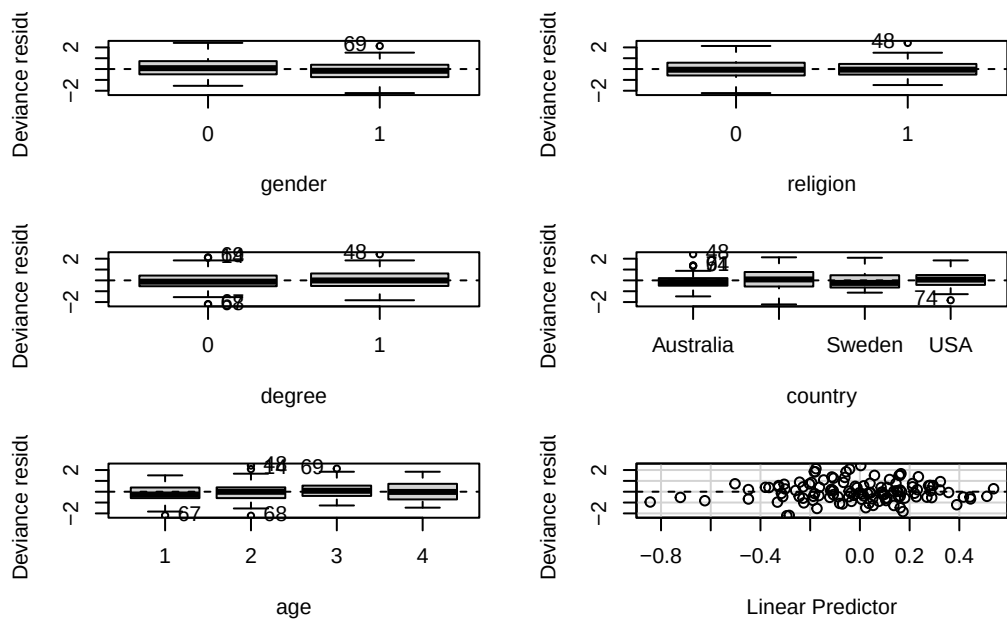
80.

```
vif(model12)
```

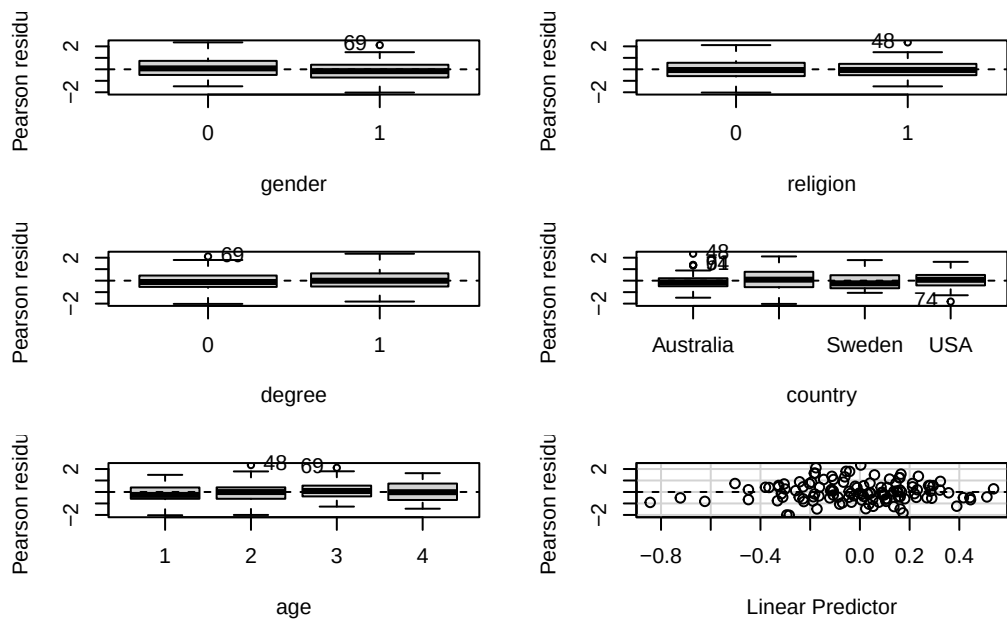
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.016862	1	1.008396
religion	2.377210	1	1.541820
degree	11.388439	1	3.374676
country	6198.983429	3	4.286012
age	23.749588	3	1.695415
religion:degree	8.434447	1	2.904212
religion:country	3102.821979	3	3.819086
degree:country	16.528969	3	1.596030
country:age	984.266463	9	1.466507

81.

```
residualPlots(model12, type = "deviance", smooth=FALSE)
```



```
residualPlots(model12, type = "pearson", smooth=FALSE)
```

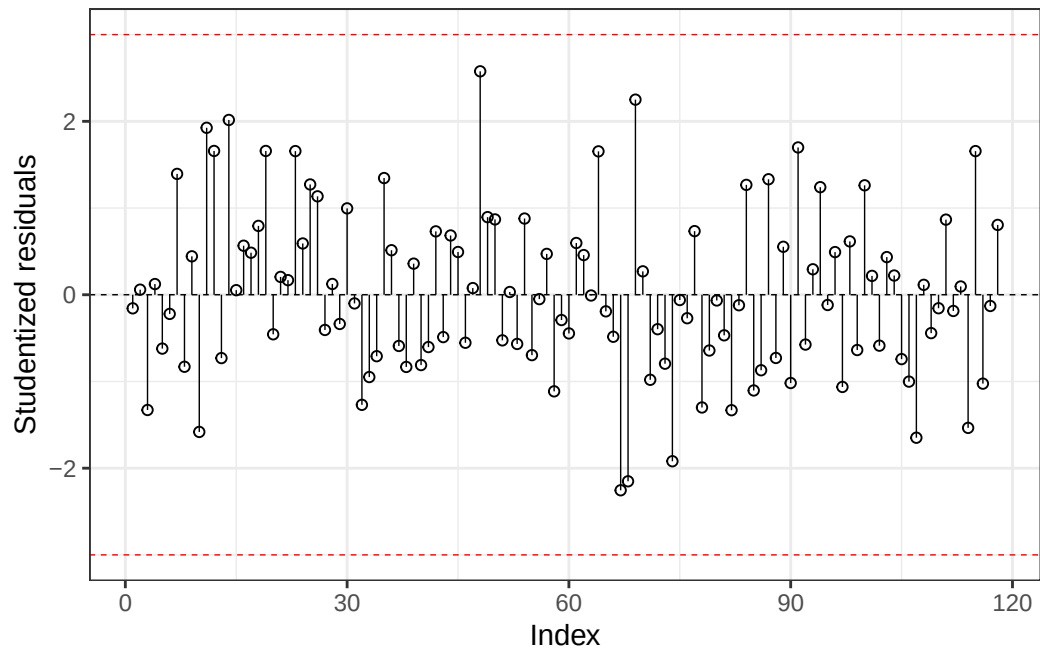


82.

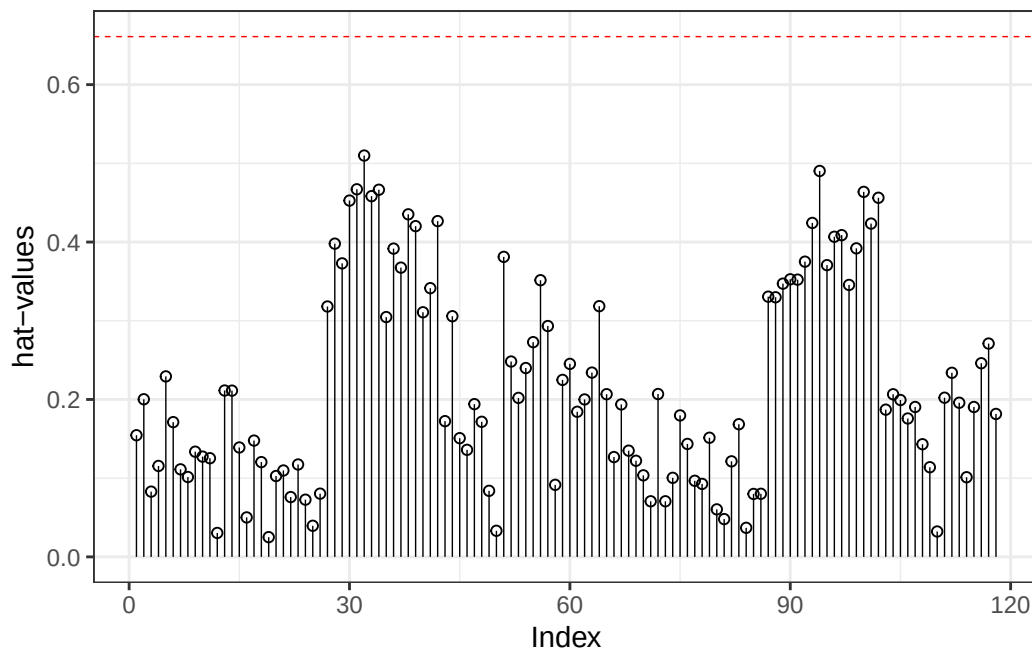
```
p <- length(coef(model12))
n <- nobs(model12)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model12_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model12),
  hat = hatvalues(model12),
  cooks = cooks.distance(model12)
)

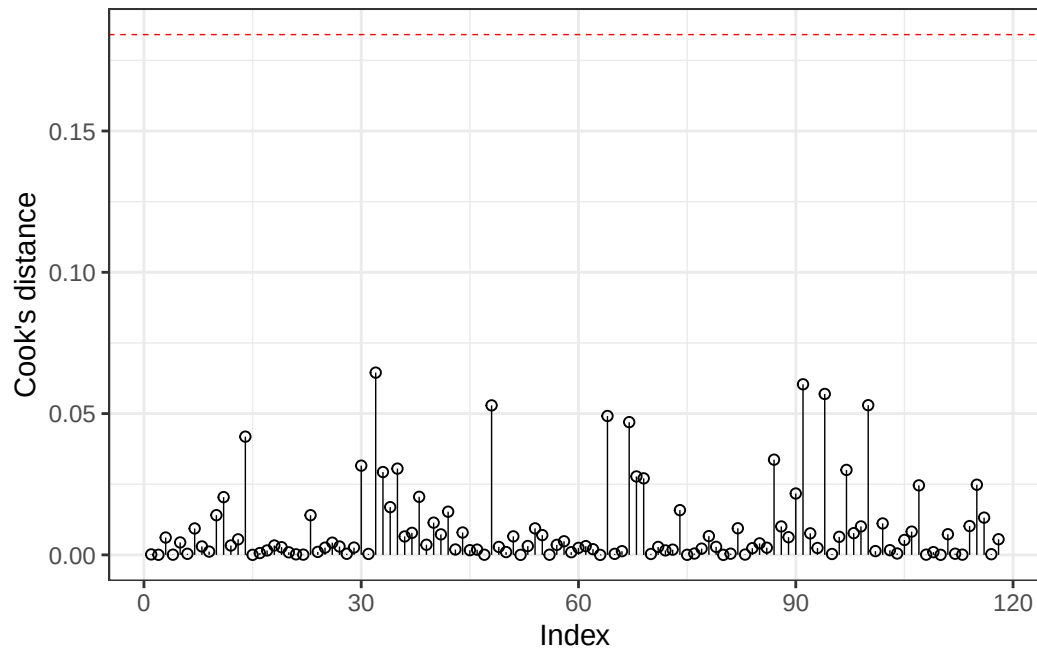
model12_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Studentized residuals") +
  theme_bw()
```



```
model12_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model12_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model12_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

83

```
check_overdispersion(model12)
```

```
# Overdispersion test
```

```
dispersion ratio = 6197.628
Pearson's Chi-Squared = 570181.745
p-value = < 0.001
```

Models 13-14: cloglog, manually reduced variables

84.

```
model6_lrt <- Anova(model6)
model6_lrt_p <- model6_lrt[, "Pr(>Chisq)"]

cbind(model6_lrt, p.adj = p.adjust(model6_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR	Chisq	Df	Pr(>Chisq)	p.adj
religion	0.8038904	1	0.36993274	0.36993274	
gender:degree	2.8487348	1	0.09144588	0.10059046	
religion:age	7.0047723	3	0.07174582	0.08768933	

85.

```
model13 <- update(model6, . ~ . - gender:degree)
summary(model13)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
    age + religion:degree + religion:country + religion:age +
    degree:country + country:age, family = binomial(link = "cloglog"),
    data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.20606	0.11487	-1.794	0.072831	.
gender1	-0.13539	0.03983	-3.399	0.000677	***
religion1	0.28100	0.12148	2.313	0.020716	*
degree1	0.23032	0.15880	1.450	0.146955	
countryNorway	-0.52523	0.19387	-2.709	0.006745	**
countrySweden	-0.35645	0.40252	-0.886	0.375871	
countryUSA	-0.07525	0.15211	-0.495	0.620783	
age2	-0.12240	0.14165	-0.864	0.387538	
age3	-0.03754	0.16262	-0.231	0.817436	
age4	-0.33197	0.19295	-1.720	0.085346	.

religion1:degree1	-0.33192	0.14875	-2.231	0.025657	*
religion1:countryNorway	0.24188	0.17628	1.372	0.170012	
religion1:countrySweden	0.49303	0.39600	1.245	0.213125	
religion1:countryUSA	-0.33673	0.13242	-2.543	0.010996	*
religion1:age2	-0.24352	0.14712	-1.655	0.097876	.
religion1:age3	-0.44164	0.16580	-2.664	0.007729	**
religion1:age4	-0.26782	0.19336	-1.385	0.166023	
degree1:countryNorway	0.14391	0.15705	0.916	0.359494	
degree1:countrySweden	-0.28200	0.16446	-1.715	0.086392	.
degree1:countryUSA	-0.17572	0.15995	-1.099	0.271930	
countryNorway:age2	0.33254	0.14170	2.347	0.018936	*
countrySweden:age2	0.27082	0.14543	1.862	0.062570	.
countryUSA:age2	0.20276	0.14571	1.392	0.164061	
countryNorway:age3	0.55356	0.15335	3.610	0.000306	***
countrySweden:age3	0.37469	0.15354	2.440	0.014674	*
countryUSA:age3	0.26489	0.16204	1.635	0.102116	
countryNorway:age4	0.55216	0.16425	3.362	0.000775	***
countrySweden:age4	0.32573	0.16614	1.961	0.049928	*
countryUSA:age4	-0.07536	0.16692	-0.451	0.651654	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 90.686 on 89 degrees of freedom
 AIC: 553.78

Number of Fisher Scoring iterations: 5

86.

```
anova(model6, model13)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
 gender:degree + religion:degree + religion:country + religion:age +
 degree:country + country:age
 Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +

```

    religion:degree + religion:country + religion:age + degree:country +
    country:age
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      88      87.838
2      89      90.686 -1  -2.8487  0.09145 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

87.

```

model13_lrt <- Anova(model13)
model13_lrt_p <- model13_lrt[, "Pr(>Chisq)"]

cbind(model13_lrt, p.adj = p.adjust(model13_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

	LR Chisq	Df	Pr(>Chisq)	p.adj
religion	0.7987898	1	0.37145545	0.37145545
religion:age	7.3279320	3	0.06214841	0.06905379

88.

```

model14 <- update(model13, . ~ . - religion:age)
summary(model14)

```

Call:

```

glm(formula = cbind(yes, no) ~ gender + religion + degree + country +
  age + religion:degree + religion:country + degree:country +
  country:age, family = binomial(link = "cloglog"), data = WVS)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.05601	0.09315	-0.601	0.547622
gender1	-0.13507	0.03981	-3.393	0.000692 ***
religion1	0.08281	0.08621	0.961	0.336728
degree1	0.24403	0.15864	1.538	0.123987
countryNorway	-0.48620	0.19092	-2.547	0.010879 *

countrySweden	-0.26028	0.40008	-0.651	0.515321	
countryUSA	-0.06116	0.15011	-0.407	0.683680	
age2	-0.30842	0.08492	-3.632	0.000282	***
age3	-0.38816	0.09551	-4.064	4.82e-05	***
age4	-0.54391	0.09976	-5.452	4.97e-08	***
religion1:degree1	-0.33985	0.14850	-2.289	0.022103	*
religion1:countryNorway	0.22865	0.17623	1.297	0.194471	
religion1:countrySweden	0.44138	0.39556	1.116	0.264494	
religion1:countryUSA	-0.37273	0.13177	-2.829	0.004676	**
degree1:countryNorway	0.14282	0.15705	0.909	0.363157	
degree1:countrySweden	-0.28774	0.16449	-1.749	0.080238	.
degree1:countryUSA	-0.17371	0.15992	-1.086	0.277381	
countryNorway:age2	0.29874	0.14006	2.133	0.032929	*
countrySweden:age2	0.21669	0.14176	1.529	0.126354	
countryUSA:age2	0.21504	0.14569	1.476	0.139944	
countryNorway:age3	0.49831	0.15211	3.276	0.001053	**
countrySweden:age3	0.29073	0.15023	1.935	0.052966	.
countryUSA:age3	0.29516	0.16180	1.824	0.068121	.
countryNorway:age4	0.52053	0.16361	3.181	0.001465	**
countrySweden:age4	0.27340	0.16383	1.669	0.095160	.
countryUSA:age4	-0.05680	0.16670	-0.341	0.733321	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 98.014 on 92 degrees of freedom
 AIC: 555.11

Number of Fisher Scoring iterations: 5

89.

```
anova(model13, model14)
```

Analysis of Deviance Table

Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +
 religion:degree + religion:country + religion:age + degree:country +


```

country:age
Model 2: cbind(yes, no) ~ gender + religion + degree + country + age +
  religion:degree + religion:country + degree:country + country:age
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      89      90.686
2      92      98.014 -3  -7.3279  0.06215 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

90.

```

model14_lrt <- Anova(model14)
model14_lrt_p <- model14_lrt[, "Pr(>Chisq)"]

cbind(model14_lrt, p.adj = p.adjust(model14_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

```

      LR Chisq Df Pr(>Chisq)      p.adj
religion 0.7987898  1  0.3714554 0.3714554

```

91.

```

pchisq(deviance(model14), df.residual(model14), lower.tail = FALSE)

```

```
[1] 0.314552
```

92.

```

model14_df = data_frame()

WVS$model14_preds <- fitted(model14, type="response")
model14_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model14_preds,
                        pred_no = WVS$total * (1 - WVS$model14_preds))

sum(model14_df$pred_yes < 5)

```

```
[1] 35
```

```
sum(model14_df$pred_yes < 1)
```

```
[1] 7
```

```
sum(model14_df$pred_no < 5)
```

```
[1] 32
```

```
sum(model14_df$pred_no < 1)
```

```
[1] 6
```

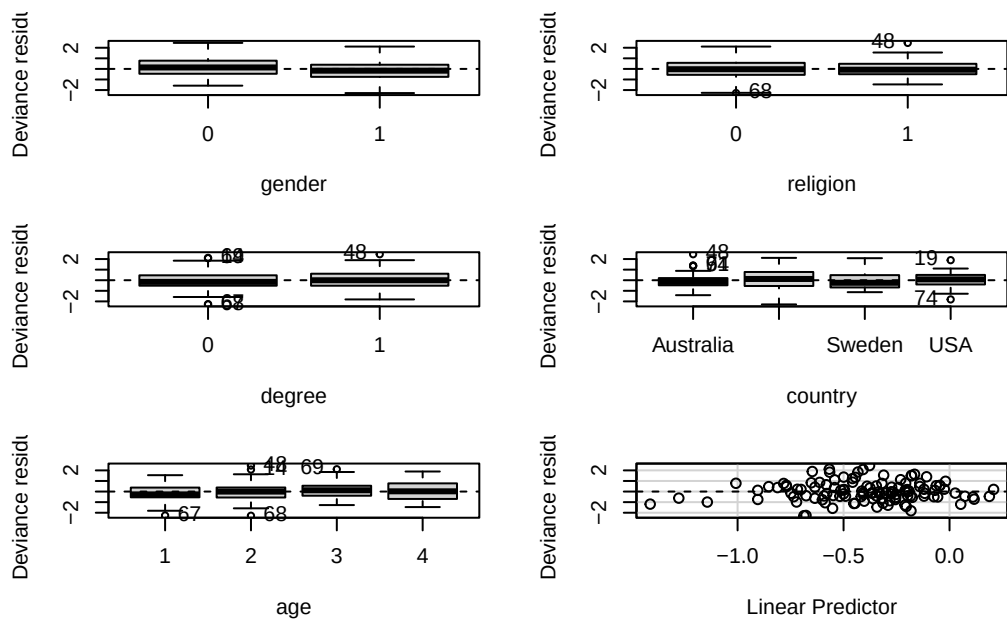
93.

```
vif(model14)
```

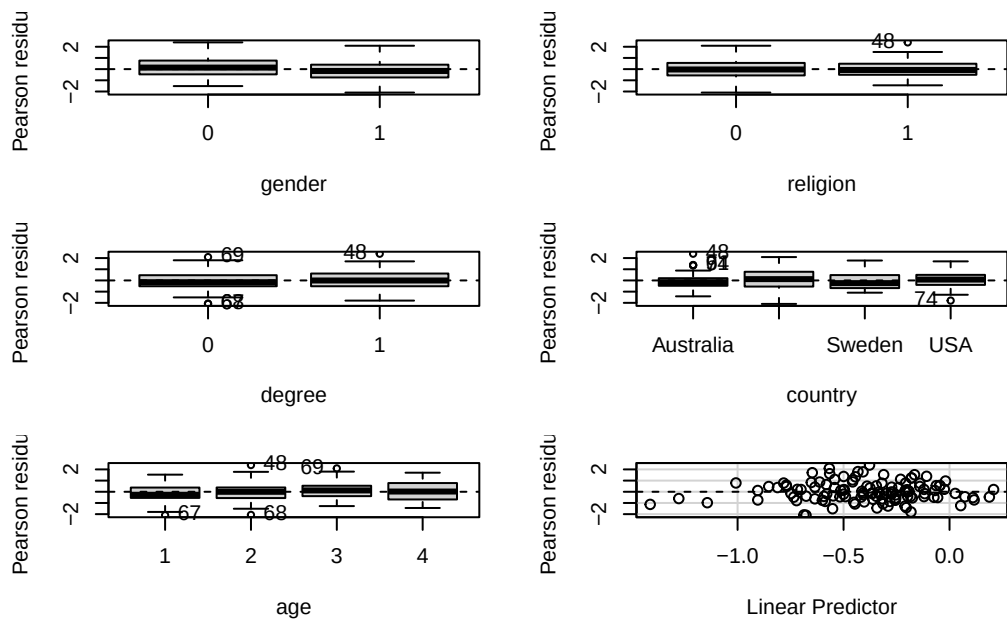
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.016965	1	1.008447
religion	2.380612	1	1.542923
degree	10.571891	1	3.251444
country	7057.069820	3	4.379630
age	24.357326	3	1.702570
religion:degree	8.129594	1	2.851244
religion:country	3584.228002	3	3.912004
degree:country	15.322117	3	1.575989
country:age	862.302006	9	1.455768

94.

```
residualPlots(model14, type = "deviance", smooth=FALSE)
```



```
residualPlots(model14, type = "pearson", smooth=FALSE)
```

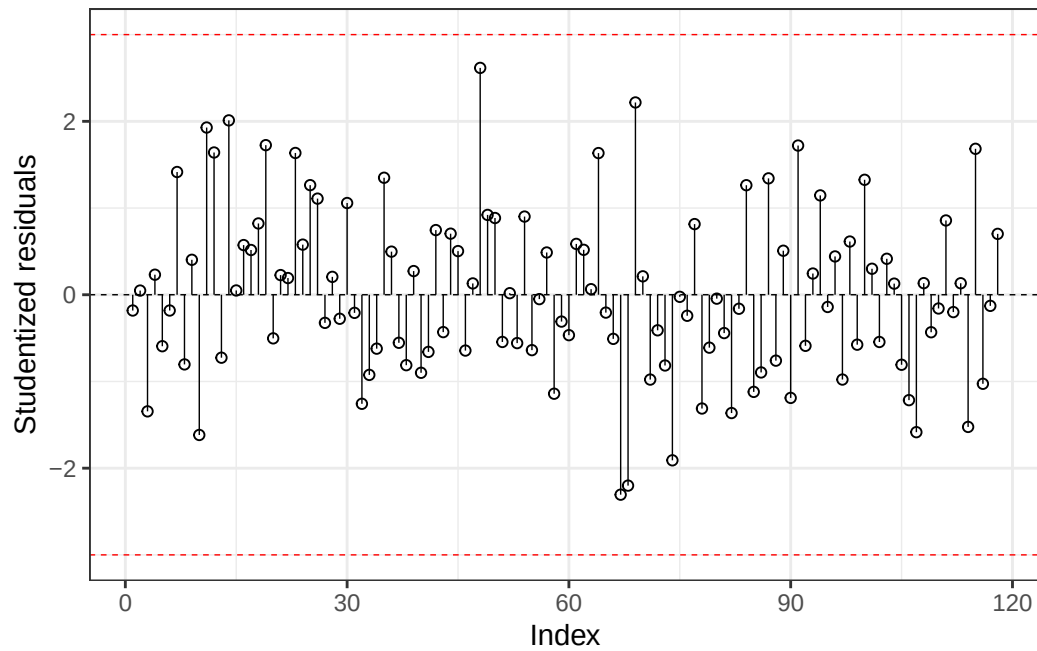


95.

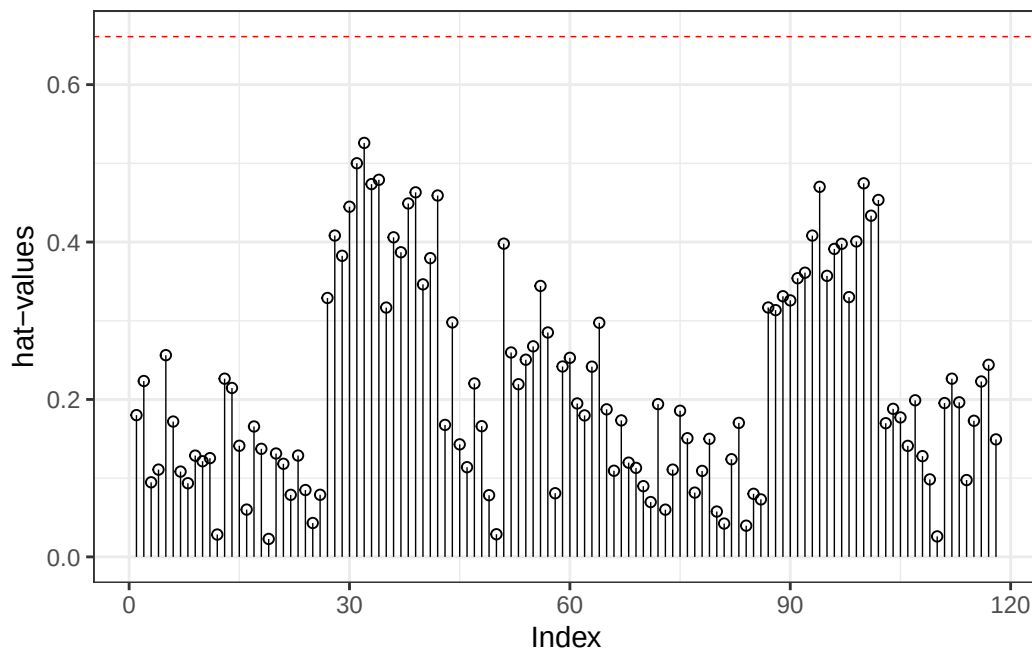
```
p <- length(coef(model14))
n <- nobs(model14)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model14_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model14),
  hat = hatvalues(model14),
  cooks = cooks.distance(model14)
)

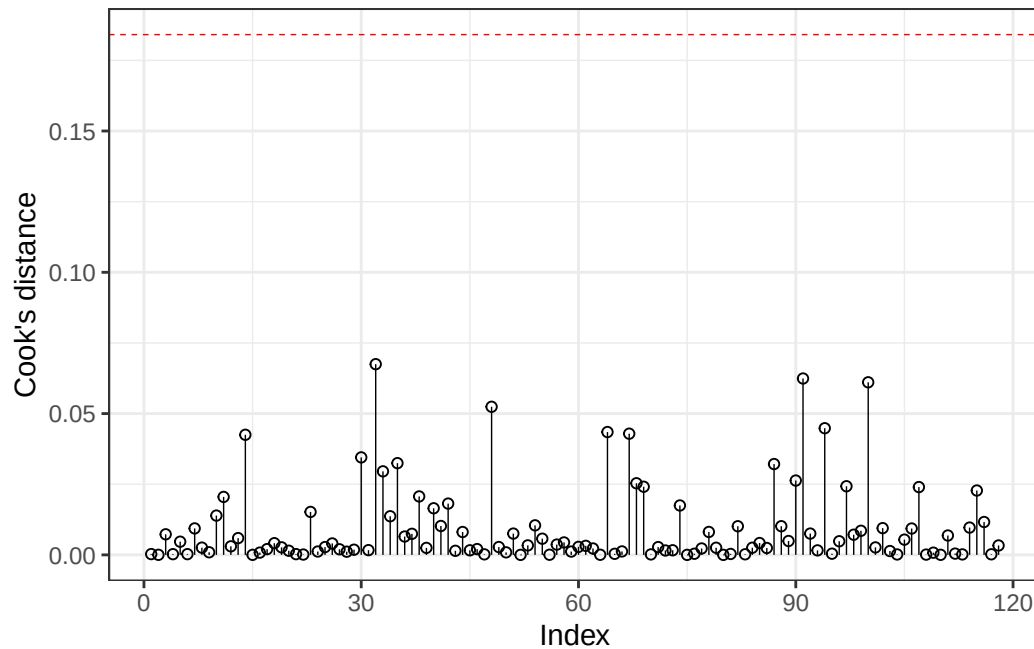
model14_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Studentized residuals") +
  theme_bw()
```



```
model14_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```



```
model14_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model14_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

96.

```
check_overdispersion(model14)
```

```
# Overdispersion test
```

```
dispersion ratio = 6186.934
Pearson's Chi-Squared = 569197.888
p-value = < 0.001
```

Models 15-16: loglog, manually reduced variables

97.

```
model8_lrt <- Anova(model8)
model8_lrt_p <- model8_lrt[, "Pr(>Chisq)"]

cbind(model8_lrt, p.adj = p.adjust(model8_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)
```

	LR Chisq	Df	Pr(>Chisq)	p.adj
religion	1.956031	1	0.1619384	0.16193843
gender:degree	2.309645	1	0.1285734	0.14143077
religion:age	7.289787	3	0.0632130	0.07726033

98.

```
model15 <- update(model8, . ~ . - gender:degree)
summary(model15)
```

Call:

```
glm(formula = cbind(no, yes) ~ gender + religion + degree + country +
    age + religion:degree + religion:country + religion:age +
    degree:country + country:age, family = binomial(link = "cloglog"),
    data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.55335	0.12079	-4.581	4.62e-06	***
gender1	0.14329	0.04005	3.578	0.000347	***
religion1	-0.30561	0.12861	-2.376	0.017493	*
degree1	-0.24217	0.17026	-1.422	0.154914	
countryNorway	0.58861	0.18321	3.213	0.001315	**
countrySweden	0.39394	0.38527	1.023	0.306544	
countryUSA	0.11770	0.15359	0.766	0.443501	
age2	0.14807	0.14460	1.024	0.305846	
age3	0.05496	0.17023	0.323	0.746790	
age4	0.36313	0.17570	2.067	0.038754	*

religion1:degree1	0.31642	0.15205	2.081	0.037425	*
religion1:countryNorway	-0.26282	0.16800	-1.564	0.117713	
religion1:countrySweden	-0.57459	0.37625	-1.527	0.126727	
religion1:countryUSA	0.33512	0.13130	2.552	0.010702	*
religion1:age2	0.25960	0.14856	1.747	0.080553	.
religion1:age3	0.46564	0.17257	2.698	0.006971	**
religion1:age4	0.26117	0.17462	1.496	0.134749	
degree1:countryNorway	-0.12195	0.16542	-0.737	0.460982	
degree1:countrySweden	0.36821	0.17481	2.106	0.035173	*
degree1:countryUSA	0.16488	0.15602	1.057	0.290628	
countryNorway:age2	-0.36545	0.15116	-2.418	0.015623	*
countrySweden:age2	-0.27863	0.17455	-1.596	0.110419	
countryUSA:age2	-0.25317	0.14593	-1.735	0.082765	.
countryNorway:age3	-0.60503	0.16507	-3.665	0.000247	***
countrySweden:age3	-0.38820	0.18030	-2.153	0.031309	*
countryUSA:age3	-0.33376	0.15881	-2.102	0.035581	*
countryNorway:age4	-0.58057	0.16983	-3.419	0.000630	***
countrySweden:age4	-0.28556	0.18827	-1.517	0.129325	
countryUSA:age4	-0.06056	0.15065	-0.402	0.687673	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.31 on 117 degrees of freedom
 Residual deviance: 86.85 on 89 degrees of freedom
 AIC: 549.94

Number of Fisher Scoring iterations: 5

99.

```
anova(model8, model15)
```

Analysis of Deviance Table

Model 1: cbind(no, yes) ~ gender + religion + degree + country + age +
 gender:degree + religion:degree + religion:country + religion:age +
 degree:country + country:age
 Model 2: cbind(no, yes) ~ gender + religion + degree + country + age +

```

    religion:degree + religion:country + religion:age + degree:country +
    country:age
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      88      84.541
2      89      86.850 -1   -2.3096  0.1286

```

100.

```

model15_lrt <- Anova(model15)
model15_lrt_p <- model15_lrt[, "Pr(>Chisq)"]

cbind(model15_lrt, p.adj = p.adjust(model15_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

```

          LR Chisq Df Pr(>Chisq)      p.adj
religion    1.999202  1 0.15738207 0.15738207
religion:age 7.520279  3 0.05703969 0.06337743

```

101.

```

model16 <- update(model15, . ~ . - religion:age)
summary(model16)

```

Call:

```

glm(formula = cbind(no, yes) ~ gender + religion + degree + country +
  age + religion:degree + religion:country + degree:country +
  country:age, family = binomial(link = "cloglog"), data = WVS)

```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.72171    0.10206  -7.071 1.53e-12 ***
gender1         0.14213    0.04004   3.550 0.000385 ***
religion1      -0.07319    0.08647  -0.846 0.397303
degree1       -0.26602    0.17031  -1.562 0.118307
countryNorway   0.55700    0.18498   3.011 0.002603 **
countrySweden   0.28489    0.38650   0.737 0.461063
countryUSA      0.10314    0.15526   0.664 0.506518

```

age2	0.33816	0.09518	3.553	0.000381	***
age3	0.41800	0.10233	4.085	4.41e-05	***
age4	0.55800	0.10183	5.480	4.26e-08	***
religion1:degree1	0.33830	0.15150	2.233	0.025552	*
religion1:countryNorway	-0.26607	0.16750	-1.588	0.112179	
religion1:countrySweden	-0.52367	0.37657	-1.391	0.164332	
religion1:countryUSA	0.34274	0.13001	2.636	0.008380	**
degree1:countryNorway	-0.12398	0.16525	-0.750	0.453102	
degree1:countrySweden	0.36940	0.17452	2.117	0.034290	*
degree1:countryUSA	0.16230	0.15582	1.042	0.297609	
countryNorway:age2	-0.32532	0.14952	-2.176	0.029566	*
countrySweden:age2	-0.21352	0.17053	-1.252	0.210535	
countryUSA:age2	-0.24140	0.14578	-1.656	0.097753	.
countryNorway:age3	-0.55831	0.16371	-3.410	0.000649	***
countrySweden:age3	-0.29567	0.17691	-1.671	0.094656	.
countryUSA:age3	-0.32095	0.15825	-2.028	0.042551	*
countryNorway:age4	-0.54433	0.16869	-3.227	0.001252	**
countrySweden:age4	-0.22428	0.18532	-1.210	0.226190	
countryUSA:age4	-0.05244	0.15023	-0.349	0.727039	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom
 Residual deviance: 94.371 on 92 degrees of freedom
 AIC: 551.47

Number of Fisher Scoring iterations: 5

102.

```
anova(model15, model16)
```

Analysis of Deviance Table

Model 1: cbind(no, yes) ~ gender + religion + degree + country + age +
 religion:degree + religion:country + religion:age + degree:country +
 country:age
 Model 2: cbind(no, yes) ~ gender + religion + degree + country + age +

```

    religion:degree + religion:country + degree:country + country:age
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      89      86.850
2      92      94.371 -3  -7.5203  0.05704 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

103.

```

model16_lrt <- Anova(model16)
model16_lrt_p <- model16_lrt[, "Pr(>Chisq)"]

cbind(model16_lrt, p.adj = p.adjust(model16_lrt_p, method = "BH")) |>
  filter(p.adj > 0.05)

```

```

      LR Chisq Df Pr(>Chisq)    p.adj
religion 1.999202  1  0.1573821 0.1573821

```

104.

```

pchisq(deviance(model16), df.residual(model16), lower.tail = FALSE)

```

```

[1] 0.4120171

```

105.

```

model16_df = data_frame()

WVS$model16_preds <- fitted(model16, type="response")
model16_df = data_frame(yes = WVS$yes,
                        no = WVS$no,
                        pred_yes = WVS$total * WVS$model16_preds,
                        pred_no = WVS$total * (1 - WVS$model16_preds))

sum(model16_df$pred_yes < 5)

```

```
[1] 34
```

```
sum(model16_df$pred_yes < 1)
```

```
[1] 6
```

```
sum(model16_df$pred_no < 5)
```

```
[1] 35
```

```
sum(model16_df$pred_no < 1)
```

```
[1] 7
```

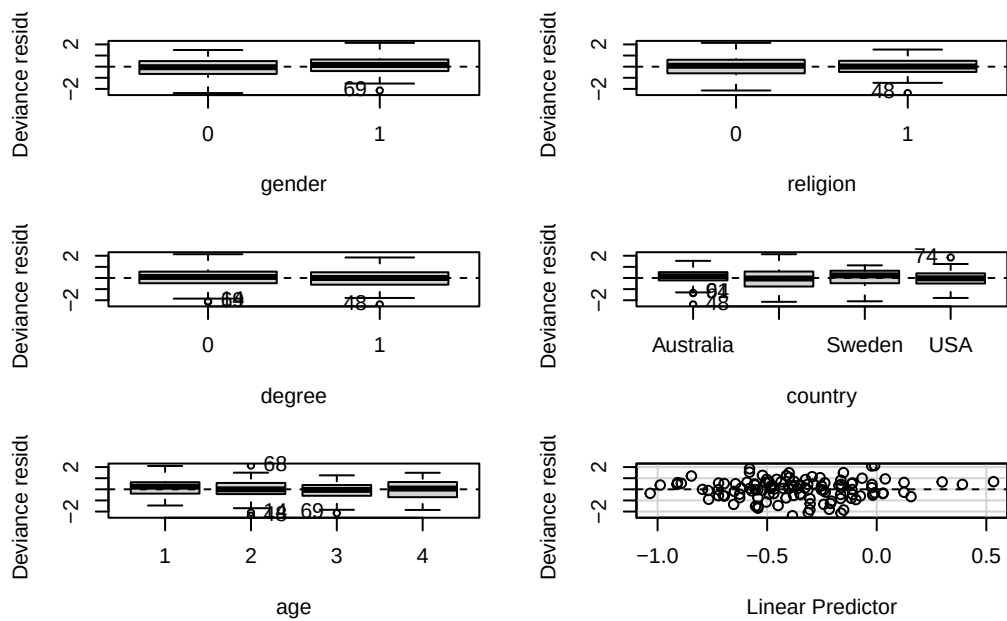
106.

```
vif(model16)
```

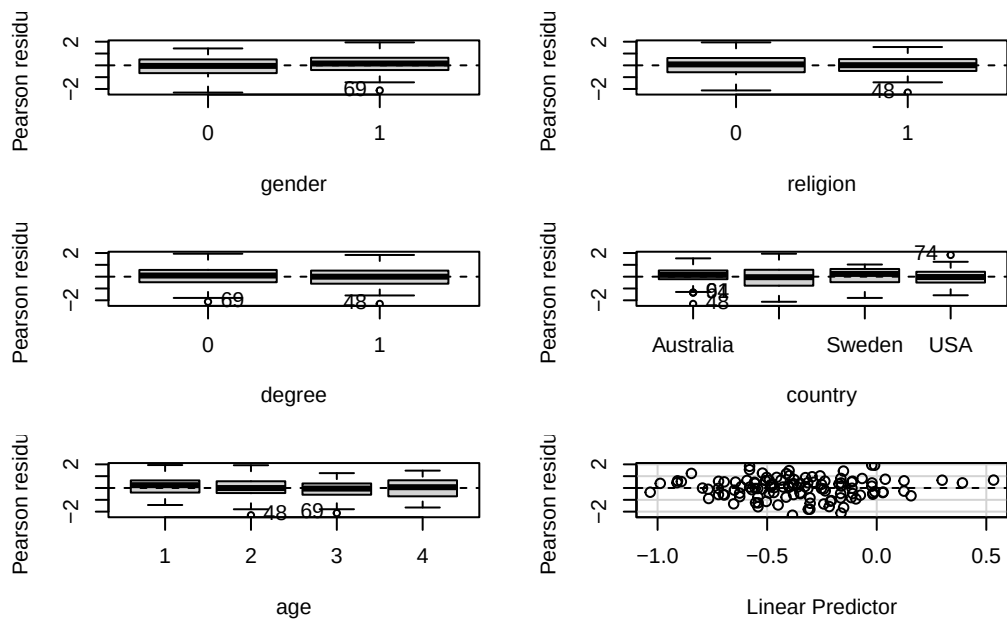
	GVIF	Df	$GVIF^{1/(2*Df)}$
gender	1.016608	1	1.008270
religion	2.391207	1	1.546353
degree	12.584015	1	3.547395
country	5420.511516	3	4.191216
age	24.776892	3	1.707423
religion:degree	9.029687	1	3.004944
religion:country	2604.916443	3	3.709360
degree:country	18.729958	3	1.629632
country:age	1206.083716	9	1.483159

107.

```
residualPlots(model16, type = "deviance", smooth=FALSE)
```



```
residualPlots(model16, type = "pearson", smooth=FALSE)
```

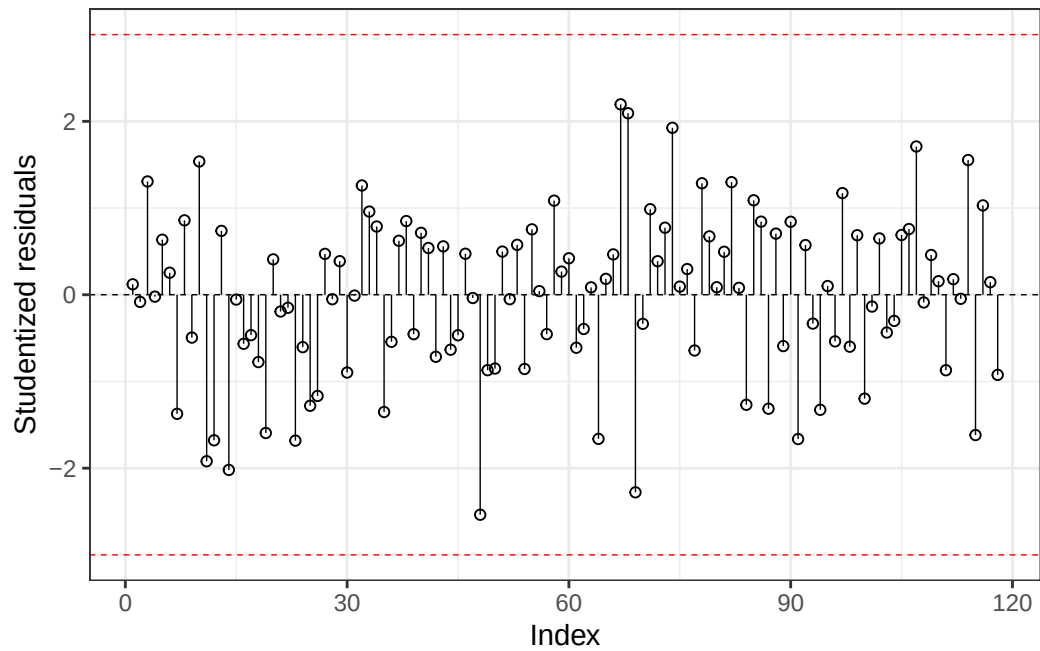


108.

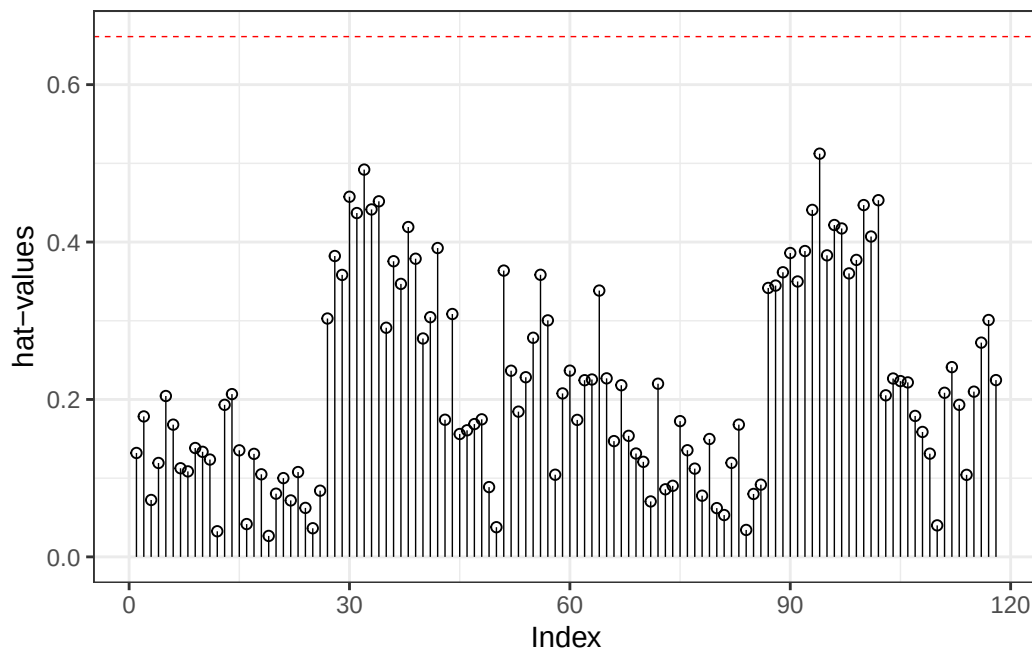
```
p <- length(coef(model16))
n <- nobs(model16)
leverage_cutoff <- (3*p)/n
cooks_cutoff <- 2/sqrt(n)

model16_inf <- data.frame(
  obs = seq(n),
  studentized = rstudent(model16),
  hat = hatvalues(model16),
  cooks = cooks.distance(model16)
)

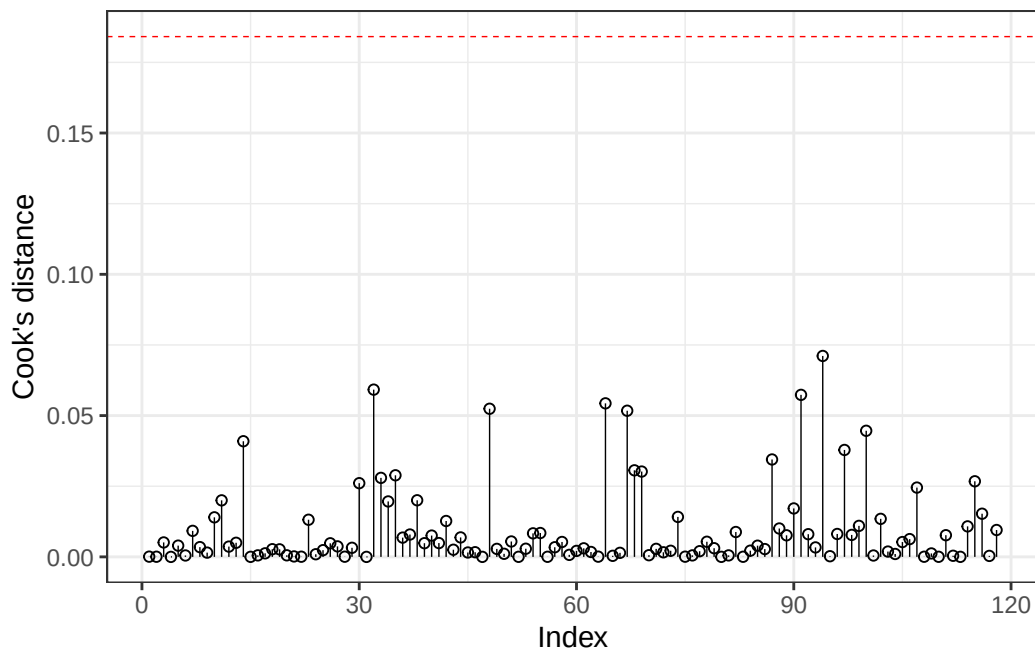
model16_inf |>
  ggplot(aes(x = obs, y = studentized)) +
  geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.25) +
  geom_hline(yintercept = -3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_hline(yintercept = 3, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_segment(aes(xend = obs, y = 0, yend = studentized), linewidth = 0.25) +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Studentized residuals") +
  theme_bw()
```



```
model16_inf |>
  ggplot(aes(x = obs, y = hat)) +
  geom_segment(aes(xend = obs, y = 0, yend = hat), linewidth = 0.25) +
  geom_hline(yintercept = leverage_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("hat-values") +
  theme_bw()
```

```
model16_inf |>
  ggplot(aes(x = obs, y = cooks)) +
  geom_segment(aes(xend = obs, y = 0, yend = cooks), linewidth = 0.25) +
  geom_hline(yintercept = cooks_cutoff, linetype = "dashed", linewidth = 0.25, color = "red") +
  geom_point(shape = 1) +
  xlab("Index") +
  ylab("Cook's distance") +
  theme_bw()
```



```
model16_inf |>
  filter(abs(studentized) > 3 |
         hat > leverage_cutoff |
         cooks > cooks_cutoff
  )
```

```
[1] obs      studentized hat      cooks
<0 rows> (or 0-length row.names)
```

109.

```
check_overdispersion(model16)
```

```
# Overdispersion test
```

```
dispersion ratio = 6205.178
Pearson's Chi-Squared = 570876.366
p-value = < 0.001
```

Model 10, 12, 14, 16 AIC comparisons

110.

```
AIC(model10)
```

```
[1] 553.4814
```

```
AIC(model12)
```

```
[1] 553.2013
```

```
AIC(model14)
```

```
[1] 555.1089
```

```
AIC(model16)
```

```
[1] 551.4653
```

Final Model: Model 10

111.

```
full_model <- glm(cbind(yes, no) ~ (gender + religion + degree + country + age)^2, family = l  
anova(model10, full_model)
```

Analysis of Deviance Table

```
Model 1: cbind(yes, no) ~ gender + religion + degree + country + age +  
  religion:degree + religion:country + degree:country + country:age  
Model 2: cbind(yes, no) ~ (gender + religion + degree + country + age)^2  
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)  
1      92      96.387  
2      78      81.316 14    15.071  0.3733
```

```
summary(model10)
```

Call:

```
glm(formula = cbind(yes, no) ~ gender + religion + degree + country +  
    age + religion:degree + religion:country + degree:country +  
    country:age, family = binomial(link = "logit"), data = WVS)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.464381	0.135927	3.416	0.000635	***
gender1	-0.194751	0.056148	-3.469	0.000523	***
religion1	0.107397	0.120573	0.891	0.373079	
degree1	0.357344	0.229829	1.555	0.119988	
countryNorway	-0.722736	0.262813	-2.750	0.005960	**
countrySweden	-0.377472	0.548017	-0.689	0.490952	
countryUSA	-0.111879	0.213022	-0.525	0.599445	
age2	-0.449527	0.125118	-3.593	0.000327	***
age3	-0.560099	0.137342	-4.078	4.54e-05	***
age4	-0.765665	0.139838	-5.475	4.37e-08	***
religion1:degree1	-0.472419	0.209384	-2.256	0.024056	*
religion1:countryNorway	0.342622	0.240680	1.424	0.154574	
religion1:countrySweden	0.671931	0.538456	1.248	0.212073	
religion1:countryUSA	-0.498727	0.182782	-2.729	0.006362	**
degree1:countryNorway	0.182686	0.224336	0.814	0.415450	
degree1:countrySweden	-0.460818	0.236069	-1.952	0.050933	.
degree1:countryUSA	-0.238187	0.220753	-1.079	0.280600	
countryNorway:age2	0.433782	0.201353	2.154	0.031214	*
countrySweden:age2	0.298672	0.218263	1.368	0.171185	
countryUSA:age2	0.316444	0.202801	1.560	0.118674	
countryNorway:age3	0.731798	0.219398	3.335	0.000852	***
countrySweden:age3	0.406447	0.228405	1.779	0.075158	.
countryUSA:age3	0.427718	0.222711	1.921	0.054794	.
countryNorway:age4	0.737769	0.230792	3.197	0.001390	**
countrySweden:age4	0.341820	0.243559	1.403	0.160487	
countryUSA:age4	-0.006884	0.220550	-0.031	0.975102	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 319.307 on 117 degrees of freedom

Residual deviance: 96.387 on 92 degrees of freedom
AIC: 553.48

Number of Fisher Scoring iterations: 4

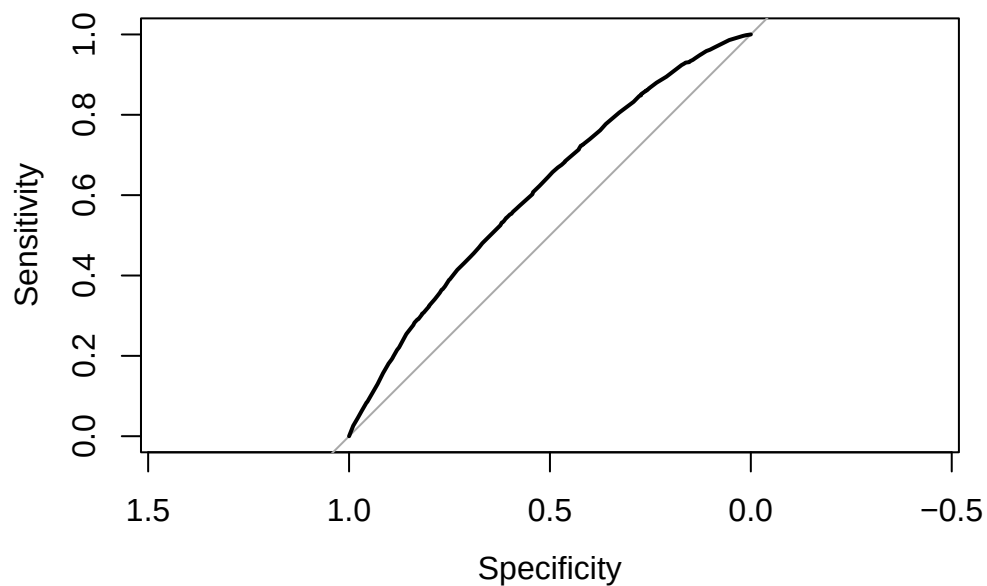
112.

```
WVS$final_fits <- fitted(model10)
```

113.

```
model10_ungrp <- glm(formula = resp ~ gender + religion + degree + country + age + religion:
WVS_ungrouped$final_fits <- fitted(model10_ungrp)

roc <- roc(data = WVS_ungrouped,
           response = resp,
           predictor = final_fits,
           auc=TRUE, plot=TRUE)
```



```
auc(roc)
```

Area under the curve: 0.6105

114.

```
WVS_ungrouped$pred_resp <- ifelse(WVS_ungrouped$final_fits > 0.5, 1, 0)

WVS_ungrouped$pred_resp <- factor(WVS_ungrouped$pred_resp, levels = c(0, 1))
WVS_ungrouped$resp <- factor(WVS_ungrouped$resp, levels = c(0, 1))

confusionMatrix(data = WVS_ungrouped$pred_resp, reference = WVS_ungrouped$resp, positive = "1")
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	1449	1063
1	1224	1645

Accuracy : 0.575
95% CI : (0.5616, 0.5882)
No Information Rate : 0.5033
P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.1496

Mcnemar's Test P-Value : 0.0008208

Sensitivity : 0.6075
Specificity : 0.5421
Pos Pred Value : 0.5734
Neg Pred Value : 0.5768
Prevalence : 0.5033
Detection Rate : 0.3057
Detection Prevalence : 0.5332
Balanced Accuracy : 0.5748

'Positive' Class : 1